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Strain-coupled one-dimensional turbulence: a single-line closure for the rapid pressure–strain and its measured limits

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The rapid distortion of turbulence by a mean strain is central to flows from wind-tunnel contractions to the stagnation regions of lifting surfaces, yet its economical prediction founders on a closure problem: the rapid pressure–strain term is a nonlocal three-dimensional spectral integral, while affordable models evolve at most one-dimensional or single-point information. We present a strain-coupled formulation of one-dimensional turbulence (ODT) and, as its analytical core, a systematic treatment of what a single line of turbulence data determines about the rapid pressure–strain term: the obstruction to a single-line closure is stated exactly, sharp bounds on the rapid term are derived from line data alone, and the closure assumption of axisymmetry about the line is not assumed but measured—it is exact for axisymmetric distortion and its plane-strain error is quantified as a function of accumulated strain. A strained-box direct numerical simulation with an exact rapid-distortion companion provides the independent spectral benchmark: it shows that the correct linear reference for line-projected spectra is broadband rather than a rigid spectral translation, and a three-way comparison localises the model's remaining deficiency in its wavenumber-uniform rapid kinematics, quantified by a distance-from-linear-theory envelope as a function of strain rapidity. Hierarchical application of the model's energy-redistributing kernel to eddy sub-scales is shown to supply the measured missing ingredient—scale-local relaxation—where it reaches the scales of the imbalance.

Key words:

1. Introduction

[M4: introduction rewrite — see markers above.]

Abstract must not spill onto p.2

2. Strain-coupled one-dimensional turbulence

The linear rapid-distortion equations, recalled below, contain two distinct ingredients: an evolution of the Fourier *amplitudes*, comprising production and the solenoidal pressure projection, and an evolution of the *wavevectors* that carries the kinematic straining of scales. The pressure part of the amplitude equation requires the full three-dimensional wavevector and is therefore not directly representable on a one-dimensional line; that incompatibility organises this section. The distortion physics is split according to its structure and each part assigned to the mechanism that can carry it faithfully: the production, which is local in physical space, is imposed as a continuous forcing of the line velocity (§ 2.3); the rapid pressure–strain, which is not, is supplied as a continuous, energy-conserving redistribution operator constructed to reproduce the homogeneous rapid-distortion evolution of the component energies under an explicit spectral closure (§ 2.4); and the wavevector kinematics are recovered by a consistent straining of the domain (§ 2.5). The slow, turbulence-driven pressure–strain and the inertial cascade remain the discrete eddy events of the standard model, reviewed first.

Throughout, $A_{ij} = \partial U_i/\partial x_j$ denotes the imposed mean velocity gradient, taken uniform across the line at each instant; the medium is incompressible, $A_{ii} = 0$. The line carries the fluctuation field $u_i(y,t)$ about this mean; $R_{ij} \equiv \overline{u_i u_j}$ is the Reynolds-stress tensor, the overline denoting the average over the statistically homogeneous directions (equivalently, the ensemble of independent realisations), and $k_t \equiv \frac{1}{2}R_{ii}$ the turbulent kinetic energy. In the homogeneous configurations used to establish and test the formulation the spatial line average and the ensemble average coincide.

2.1. *The standard model*

This subsection reviews the one-dimensional turbulence model of Kerstein (1999) in the vector-valued formulation of Kerstein *et al.* (2001), in the notation used throughout; nothing in it is original. The model evolves a three-component velocity field $u_i(y,t)$, $i = 1, 2, 3$, on a one-dimensional domain whose coordinate y is identified with the direction x_2. The fields represent individual realisations; ensemble statistics are accumulated over many independent simulations. Two interleaved mechanisms advance the state: between events the fields obey ordinary one-dimensional diffusion, $(\partial_t - \nu\,\partial_y^2)\,u_i = 0$, with kinematic viscosity ν; and a stochastic sequence of instantaneous *eddy events* represents the effect of individual turbulent overturns.

Because a continuum rotational motion cannot be represented on a line, an eddy event is an instantaneous, measure-preserving rearrangement of the profiles together with an additive velocity change,

$$u_i(y) \;\longrightarrow\; u_i\big(f(y)\big) + c_i\,K(y), \tag{2.1}$$

where f is the *triplet map* (Kerstein 1991, 1999), defined on an interval $[y_0, y_0 + l]$ of size l by

$$f(y) = y_0 + \begin{cases} 3(y - y_0), & y_0 \le y \le y_0 + \frac{1}{3}l, \\ 2l - 3(y - y_0), & y_0 + \frac{1}{3}l \le y \le y_0 + \frac{2}{3}l, \\ 3(y - y_0) - 2l, & y_0 + \frac{2}{3}l \le y \le y_0 + l, \\ y - y_0, & \text{otherwise}, \end{cases} \tag{2.2}$$

written through its inverse, so that fluid at $f(y)$ is placed at y: the map compresses the interval to one third of its length, inserts three copies, reverses the central copy, and leaves the exterior unchanged. It is *measure preserving* (the non-local analogue of a divergence-free rearrangement), *continuous*, and *scale local* (it changes gradients by at most an order-unity factor) (Kerstein *et al.* 2001)—the three invariants whose preservation under an imposed mean strain is established in § 2.3. The additive term is built from the *kernel*

$$K(y) \equiv y - f(y), \qquad \int K(y)\,\mathrm{d}y = 0, \qquad \int K^2(y)\,\mathrm{d}y = \tfrac{4}{27}\,l^3, \tag{2.3}$$

non-zero only within the eddy interval; the first identity guarantees that the additive change conserves the momentum of each component, and the second fixes the energetics of the redistribution.

The kernel term is the model surrogate for the pressure-mediated redistribution of energy among velocity components (Wunsch & Kerstein 2001; Kerstein *et al.* 2001). Requiring that an event redistribute kinetic energy among components while conserving the total, with a rule invariant under permutation of the component indices, gives

$$\Delta E_i = \alpha \sum_j T_{ij}\, Q_j, \qquad \mathsf{T} = \frac{1}{2}\begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \qquad Q_i \equiv \tfrac{27}{8}\,\rho_0\, l\, u_{i,K}^2, \tag{2.4}$$

where $u_{i,K} \equiv l^{-2}\int u_i(f(y))\,K(y)\,\mathrm{d}y$, Q_i is the available kinetic energy of component i within the eddy, ρ_0 the constant density, and $\alpha \in [0, 1]$ the parameter setting the fraction of extractable energy transferred towards isotropy. Solving the per-component energy balance for the amplitudes yields

$$c_i = \frac{27}{4l}\left[-u_{i,K} + \mathrm{sgn}(u_{i,K})\sqrt{u_{i,K}^2 + \alpha \sum_j T_{ij}\, u_{j,K}^2}\,\right], \tag{2.5}$$

real for $0 \le \alpha \le 1$. The transfer matrix T equalises the component energies: the construction is a model of the return-to-isotropy (*slow*) part of the pressure–strain interaction, a correspondence made precise in § 2.2. The standard model contains no counterpart of the rapid, mean-strain-driven part.

The event sequence is a Poisson process whose rate reflects the instantaneous flow state: each candidate eddy (y_0, l) is assigned a time scale τ through $(l/\tau)^2 \sim u_{2,K}^2 + \alpha \sum_j T_{2j} u_{j,K}^2 - Z\,\nu^2/l^2$, the last term a viscous penalty with order-unity parameter Z, and the event-rate density is $\lambda(y_0, l; t) = C/\big(l^2\tau\big)$ with rate constant C; candidates with negative discriminant are forbidden. The inertial-range cascade and the $k^{-5/3}$ spectrum are *outcomes* of repeated sampling from this rate, not inputs (Kerstein 1999); C, Z and a large-eddy suppression parameter are the only tunable constants (Stephens & Lignell 2021). Finally, one diagnostic caveat of Kerstein *et al.* (2001) is respected throughout: because the line velocity does not itself advect fluid, off-diagonal products of line velocity components are not momentum fluxes; diagonal component spectra and variances are read from the fields, whereas shear stresses are read from eddy-induced transport.

2.2. *Rapid and slow pressure–strain, and linear rapid distortion*

The Reynolds-stress transport equation for a fluctuating field about a mean with constant gradient A_{ij} reads, in homogeneous turbulence,

$$\frac{\mathrm{d}R_{ij}}{\mathrm{d}t} = \mathcal{P}_{ij} + \Pi_{ij} - \varepsilon_{ij}, \qquad \mathcal{P}_{ij} = -A_{ik}R_{kj} - A_{jk}R_{ki}, \qquad \Pi_{ij} = \left\langle \tfrac{p'}{\rho}\big(\partial_j u_i' + \partial_i u_j'\big)\right\rangle, \tag{2.6}$$

with ε_{ij} the dissipation tensor (Pope 2000). Taking the divergence of the fluctuating momentum equation gives the pressure as the solution of a Poisson equation whose source splits into a part linear in the mean gradient and a part quadratic in the fluctuations,

$$\frac{1}{\rho}\nabla^2 p' = \underbrace{-2\,A_{lm}\,\partial_l u'_m}_{\text{rapid}} \;-\; \underbrace{\partial_l\partial_m\big(u'_l u'_m - \langle u'_l u'_m\rangle\big)}_{\text{slow}}, \tag{2.7}$$

and the pressure–strain correlation inherits the decomposition $\Pi_{ij} = \Pi^{(\mathrm{r})}_{ij} + \Pi^{(\mathrm{s})}_{ij}$ into a *rapid* part driven by the mean strain and a *slow* part driven by the turbulence alone. The slow part relaxes the Reynolds stress towards isotropy—the mechanism that the kernel (2.4)–(2.5) reproduces in model form. The rapid part is linear in A_{lm} and is determined by the velocity spectrum tensor $\Phi_{ij}(\boldsymbol{\kappa})$; the standard model represents neither $\mathcal{P}_{ij}$ nor $\Pi^{(\mathrm{r})}_{ij}$.

When the total strain accumulates in a time short compared with the eddy turnover time, the response is computed exactly by linear theory (Batchelor & Proudman 1954; Hunt 1973; Hunt & Carruthers 1990). Writing the fluctuation as Fourier modes that follow the mean flow, each mode of time-dependent wavevector $\boldsymbol{\kappa}(t)$ and amplitude $\hat{u}_i$ evolves, in the inviscid rapid limit, according to

$$\frac{\mathrm{d}\kappa_i}{\mathrm{d}t} = -A_{ji}\,\kappa_j, \qquad \frac{\mathrm{d}\hat{u}_i}{\mathrm{d}t} = \left(\frac{2\,\kappa_i\kappa_m}{\kappa^2} - \delta_{im}\right) A_{mn}\,\hat{u}_n. \tag{2.8}$$

The bracketed operator combines the production $-A_{im}\hat{u}_m$ with the rapid pressure response, the solenoidal projection $P_{ij} = \delta_{ij} - \kappa_i\kappa_j/\kappa^2$ applied to the strained field; the factor of two is fixed by continuity, since only that coefficient makes $\mathrm{d}(\kappa_i\hat{u}_i)/\mathrm{d}t$ vanish under (2.8). Two features are central to what follows. First, the pressure response is *non-local* and intrinsically *three-dimensional*: it requires all of $\kappa_1, \kappa_2, \kappa_3$, whereas a single line resolves only the wavenumber along its own coordinate, so (2.8) cannot be evaluated on the line as written. Second, the rapid/slow decomposition furnishes the organising principle of the formulation: the slow part is already carried by the kernel, and it remains to supply the production and a rapid redistribution consistent with (2.8), expressed in quantities available on the line.

2.3. *The strain-coupled line equation and its structural admissibility*

The production term of the fluctuation momentum equation, $-A_{ij}u_j$, is a pointwise linear combination of the local velocity components and is therefore exactly representable on the line. Between eddy events the molecular evolution is replaced by

$$\partial_t u_i = \nu\,\partial_y^2 u_i \;-\; A_{ij}\,u_j \;+\; B_{ij}\,u_j, \tag{2.9}$$

in which $-A_{ij}u_j$ is the production and $B_{ij}u_j$ is the rapid pressure–strain operator constructed in § 2.4; the eddy events (2.1)–(2.5) are retained without modification, and the advancement is split between the continuous phase (2.9) and the discrete event sequence.

The forcing must not violate the structural properties on which the model rests, and it does not; we record the argument as a consistency check. The measure preservation, scale locality and kernel identities (2.3) are geometric properties of the map f and kernel K on a fixed domain, and the operator splitting applies the discrete event exactly as in the standard model, so they hold verbatim. Continuity of the fields is preserved because, for a mean gradient uniform along the line, the operator $\mathcal{A}_{ij} \equiv -A_{ij} + B_{ij}$ acts pointwise and the solution of the non-diffusive part of (2.9) over a sub-step is

$$u_i(y,t) = \big[\exp\!\int_{t_0}^{t} \mathcal{A}\,\mathrm{d}t'\big]_{ij}\,u_j(y,t_0), \tag{2.10}$$

a y-independent matrix, which maps continuous profiles to continuous profiles while the molecular term is smoothing. The forcing displaces no fluid along y, so the Lebesgue measure of the domain is untouched; the deliberate straining of the domain is a separate operation (§ 2.5).

The energetic role of the forcing is exact. At fixed domain length the production term evolves the Reynolds stress as

$$\left.\frac{\mathrm{d}R_{ij}}{\mathrm{d}t}\right|_{\mathrm{prod}} = -A_{ik}R_{kj} - A_{jk}R_{ki} \equiv \mathcal{P}_{ij}, \tag{2.11}$$

exactly the Reynolds-stress production tensor of (2.6): the term carries no model assumption. The kinetic-energy budget of the formulation thus separates into an exact production by the mean strain, energy-conserving redistribution (the kernels), and dissipation by molecular diffusion—the structure of the exact transport equation. The forcing adds the one term the standard model lacks without disturbing the terms it already represents.

2.4. *The rapid redistribution operator*

Were the continuous evolution to consist of production and diffusion only ($B_{ij} = 0$), the line would reproduce $\mathrm{d}R_{ij}/\mathrm{d}t = \mathcal{P}_{ij}$, whereas the exact inviscid rapid-distortion balance is $\mathrm{d}R_{ij}/\mathrm{d}t = \mathcal{P}_{ij} + \Pi^{(\mathrm{r})}_{ij}$. The omission is not small: for turbulence isotropic at the onset of straining the classical result derived below gives $\Pi^{(\mathrm{r})}_{ij} = -\frac{3}{5}\mathcal{P}_{ij}$, so production acting alone overstates the initial distortion rate by a factor of $\frac{5}{2}$; under sustained strain it also lets a component energy grow without the bound that incompressibility imposes. Representing $\Pi^{(\mathrm{r})}_{ij}$ is essential, not optional.

2.4.1. *The exact rapid pressure–strain and the closure problem*

The exact rapid term follows from the amplitude equation. Writing (2.8) as $\mathrm{d}\hat{u}_i/\mathrm{d}t = M_{in}\hat{u}_n$ with $M_{in} = -A_{in} + 2(\kappa_i\kappa_m/\kappa^2)A_{mn}$, and noting that the incompressible wavevector dynamics preserve the measure $\mathrm{d}\boldsymbol{\kappa}$ (their phase-space divergence is $-A_{ii} = 0$), the spectrum tensor $\Phi_{ij} = \langle \hat{u}_i^*\hat{u}_j\rangle$ yields

$$\frac{\mathrm{d}R_{ij}}{\mathrm{d}t} = \int \left(M_{in}\Phi_{nj} + M_{jn}\Phi_{in}\right)\mathrm{d}\boldsymbol{\kappa} = \mathcal{P}_{ij} + \Pi^{(\mathrm{r})}_{ij}, \tag{2.12}$$

in which the production separates from the pressure contribution,

$$\Pi^{(\mathrm{r})}_{ij} = 2A_{mn}\left(\mathcal{R}_{ijmn} + \mathcal{R}_{jimn}\right), \qquad \mathcal{R}_{ijmn} \equiv \int \frac{\kappa_i\kappa_m}{\kappa^2}\Phi_{nj}(\boldsymbol{\kappa})\,\mathrm{d}\boldsymbol{\kappa}. \tag{2.13}$$

The tensor is traceless—by solenoidality $\kappa_n\Phi_{nj} = 0$, the trace integrand contains $\kappa_i\Phi_{ni} = 0$—so the rapid term redistributes energy among components without changing the total, in keeping with the energy-conserving character of the kernel mechanism. Equation (2.13) also exposes the central difficulty: the rapid term is a functional of the spectrum tensor through the orientation-weighted integral $\mathcal{R}_{ijmn}$, not of the single-point Reynolds stress—the classical closure problem of the rapid pressure–strain, common to every single-point model (Crow 1968; Launder *et al.* 1975; Pope 2000) and intrinsic to the physics. A line carries R_{ij} and its own one-dimensional spectra, but not the distribution of energy over wavevector *orientation* that $\mathcal{R}_{ijmn}$ requires; a spectral-tensor closure is therefore unavoidable, and we make it explicit. (What a line's data *do* determine about this integral, and how sharply, is the analytical question taken up in § 5.)

2.4.2. *Baseline isotropic closure*

The minimal closure takes the spectrum tensor instantaneously isotropic, $\Phi_{nj} = \frac{E(\kappa)}{4\pi\kappa^2}(\delta_{nj} - \kappa_n\kappa_j/\kappa^2)$ with $\int E\,\mathrm{d}\kappa = k_t$. With the angular averages $\overline{e_i e_m} = \frac{1}{3}\delta_{im}$ and $\overline{e_i e_m e_n e_j} = \frac{1}{15}(\delta_{im}\delta_{nj} + \delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn})$ for $e_i = \kappa_i/\kappa$,

$$\mathcal{R}^{\mathrm{iso}}_{ijmn} = k_t\left[\tfrac{4}{15}\delta_{im}\delta_{nj} - \tfrac{1}{15}\left(\delta_{in}\delta_{mj} + \delta_{ij}\delta_{mn}\right)\right], \tag{2.14}$$

and substitution into (2.13) gives, with $S_{ij} = \frac{1}{2}(A_{ij} + A_{ji})$ and $A_{nn} = 0$,

$$\Pi^{(\mathrm{r}),\mathrm{iso}}_{ij} = \tfrac{4}{5}\,k_t\,S_{ij} = -\tfrac{3}{5}\,\mathcal{P}_{ij} \quad (\text{isotropic } R_{ij}), \tag{2.15}$$

the exact response of initially isotropic turbulence to a weak rapid strain (Crow 1968) and the content of the isotropisation-of-production model with $C_2 = \frac{3}{5}$ (Pope 2000)—obtained here, with no adjustable constant, as the isotropic limit of (2.13). For anisotropic turbulence $\mathcal{R}_{ijmn}$ must be modelled as a tensor function of R_{ij}; the general representation linear in the anisotropy is that of Launder *et al.* (1975), to which the construction reduces in form. We adopt (2.14) as the baseline and assess the anisotropic correction in § 3.

2.4.3. *Realisation on the line*

Given the closed rapid term—symmetric and traceless—we require a pointwise operator B_{ij} whose action on the line reproduces it. Demanding that the operator evolve the Reynolds stress by exactly $\Pi^{(\mathrm{r})}_{ij}$,

$$\left(\mathsf{BR} + \mathsf{RB}\right)_{ij} = \Pi^{(\mathrm{r})}_{ij}, \tag{2.16}$$

a continuous Lyapunov equation with, for positive-definite R, a unique symmetric solution B. The solution conserves kinetic energy automatically—contracting (2.16) gives $2B_{ij}R_{ij} = \Pi^{(\mathrm{r})}_{ii} = 0$, so the rate of working of the operator on the line vanishes. In the isotropic case $\mathsf{R} = \frac{2}{3}k_t\mathsf{I}$ it reduces to the closed form

$$B_{ij} = \tfrac{3}{5}\,S_{ij}, \qquad \mathcal{A}_{ij} = -A_{ij} + \tfrac{3}{5}\,S_{ij}. \tag{2.17}$$

B_{ij} is evaluated from the running Reynolds stress and applied continuously, paced by the mean strain, exactly as the production is. This is deliberate: the rapid pressure–strain is by construction the part of the redistribution driven by the mean gradient and therefore acts at the strain rate, in contrast to the slow redistribution that the eddy kernel supplies at the eddy rate. Strain-driven effects (production and rapid pressure–strain) are continuous and enter through $\mathcal{A}_{ij}$; turbulence-driven effects (the inertial cascade and the slow return to isotropy) remain the discrete events.

The sense in which the formulation is rapid-distortion consistent can now be stated precisely, again as a consistency check rather than a theorem. By (2.11) the production term contributes $\mathcal{P}_{ij}$ to $\mathrm{d}R_{ij}/\mathrm{d}t$ and by construction (2.16) the operator contributes $\Pi^{(\mathrm{r})}_{ij}$; their sum is the exact balance (2.12) for the prescribed closure, with no adjustable constant, recovering (2.15) at the onset of distortion of isotropic turbulence. The scope of the statement must be drawn as carefully as the result: the consistency holds at the level of the *component energies* and is *conditional on the spectral closure*; the isotropic closure is exact only at onset, and its accuracy for anisotropic states is that of the isotropisation-of-production approximation it reproduces. The *spectral* distribution of the distorted energy is not enforced by (2.9): the operator acts uniformly on every resolved wavenumber, and the distribution of energy across scales emerges from its interaction with diffusion, the cascade and the domain straining. It is therefore a genuine prediction, tested in §§ 3–4—where the uniform, wavenumber-blind action of B_{ij} is also identified as the model's measured spectral deficiency (§ 4.1.1).

2.5. *Wavevector kinematics by domain straining*

The second ingredient of (2.8), the wavevector evolution, carries the kinematic compression and stretching of scales and is independent of the amplitude evolution: in the method-of-characteristics reading of linear theory, the wavevectors are the characteristics along which the amplitudes evolve. On the line the relevant wavenumber is κ_2. We restrict attention to a line aligned with a principal axis of the mean strain, so that $A_{12} = A_{32} = 0$ and a material element of the line remains a coordinate line—as holds for the irrotational mean flow of a stagnation region. Then $\mathrm{d}\kappa_2/\mathrm{d}t = -A_{22}\,\kappa_2$, and since a material element of length ℓ stretches as $\dot{\ell}/\ell = A_{22}$ while wavenumbers scale as $1/\ell$, a uniform dilatation of the domain reproduces the wavevector kinematics exactly:

$$\frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t} = A_{22}, \qquad \Longrightarrow \qquad \kappa_2(t) = \kappa_2(t_0)\,\exp\Bigl(-\int_{t_0}^{t} A_{22}\,\mathrm{d}t'\Bigr), \tag{2.18}$$

with L the domain length, implemented within the Lagrangian adaptive-mesh advancement of the model (Lignell *et al.* 2013) by imposing the mean-flow stretching on the mesh.

This step is not redundant with the amplitude operator. The Reynolds stress is an intensive second moment and is invariant under the uniform dilatation: the dilatation relabels the scales that carry the energy without changing the component energies, while $\mathcal{A}_{ij}$ changes the component energies without relabelling scales. The two operations are the physical-space counterparts of the characteristic motion and the amplitude evolution in (2.8), and together they account for the mean straining without double counting. Two limitations are inherent to representing a three-dimensional kinematics on one axis: the transverse wavenumbers also strain but are not represented, their effect being subsumed in the spectral closure, which already supplies the orientation information the line lacks; and a rotational mean flow, or a line not aligned with a principal axis, tilts the material line out of the coordinate direction and lies outside the formulation.

2.6. *Summary of the formulation*

The strain-coupled model advances the line field over a sub-step by composing (a) a *continuous strain–diffusion phase on a dilating domain*—integrate $\partial_t u_i = \nu\partial_y^2 u_i + \mathcal{A}_{ij}u_j$ with $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$, the production exact and B_{ij} given by (2.16) from the running Reynolds stress and the spectral closure (baseline (2.17)), while dilating the domain by (2.18)—and (b) *discrete eddy events* sampled from the standard rate distribution: the triplet map and the slow kernel, unchanged. Step (a) carries the mean-strain physics; step (b) carries the turbulence-driven physics; standard ODT is recovered exactly when $A_{ij} = 0$.

The relative importance of the two steps is governed by the ratio of the mean strain rate $S \equiv (2S_{ij}S_{ij})^{1/2}$ to the turbulence relaxation rate,

$$\chi \equiv \frac{S\,k_t}{\varepsilon}. \tag{2.19}$$

For $\chi \gg 1$ the distortion accumulates faster than the turbulence can respond, the eddy events are negligible over the distortion time, and the formulation reduces to the rapid-distortion limit of step (a), exact for the component energies under the closure. For $\chi \ll 1$ the turbulence relaxes between increments of strain and the slow kernel dominates. The stagnation region of a leading edge is the intermediate regime $\chi = O(1)$, in which neither limit is uniformly valid and both mechanisms contribute; the fidelity of the formulation across this range, and of the emergent distorted spectrum, are assessed against rapid-distortion solutions and scale-resolving data in §§ 3–4.

3. Verification, and moment-level validation

We separate two questions. *Verification* asks whether the formulation and its implementation reproduce a known exact solution in the regime where one exists—the rapid-distortion limit, tested analytically and in the solver below. *Validation* asks whether the model reproduces the behaviour of real strained turbulence outside that limit; the spectral validation, against the strained-box DNS with its exact-RDT companion, is the subject of § 4, and this section closes with the moment-level validation against the strained-turbulence DNS of Lee & Reynolds (1985).

3.1. *Canonical strains and the exact onset rate*

Two irrotational homogeneous strains admit exact rapid-distortion solutions and isolate the physics of interest. *Plane strain*,

$$\mathsf{A} = \mathrm{diag}(a, -a, 0), \qquad a > 0, \tag{3.1}$$

is the strain of a two-dimensional stagnation region: with x_1 tangential to the surface, x_2 normal to it and x_3 spanwise, the component compressed by the strain is the wall-normal (upwash) component u_2, whose amplification is the pre-impact distortion relevant to leading-edge noise. *Axisymmetric strain*,

$$\mathsf{A} = \mathrm{diag}(-\tfrac{1}{2}\gamma, -\tfrac{1}{2}\gamma, \gamma), \qquad \gamma > 0, \tag{3.2}$$

is the strain of a contraction or a three-dimensional stagnation flow and provides an independent test in which two components are amplified and one suppressed. For both, A is symmetric, so the principal-axis alignment assumed in § 2.5 holds; the line is taken along x_2.

For turbulence isotropic at the onset of straining the continuous evolution (2.9) gives, by (2.11) and (2.15),

$$\left.\frac{\mathrm{d}R_{ij}}{\mathrm{d}t}\right|_0 = \mathcal{P}_{ij} + \Pi^{(\mathrm{r}),\mathrm{iso}}_{ij} = \tfrac{2}{5}\,\mathcal{P}_{ij} = -\tfrac{8}{15}\,k_t\,S_{ij}, \tag{3.3}$$

the exact rapid-distortion rate for initially isotropic turbulence, with no adjustable constant. This is the sharpest verification available: a closed-form prediction the model must satisfy identically, not approximately. Evaluated for the two canonical strains it gives the component rates of table 1; for plane strain the predicted upwash growth rate $+\frac{8}{15}k_t a$ is to be compared with the $\frac{4}{3}k_t a$ that production alone would give—the factor-$\frac{5}{2}$ overstatement anticipated in § 2.4. A model that passes (3.3) for both strains has its production and rapid terms verified jointly; failure would indicate an error in the operator $\mathcal{A}_{ij}$ or its implementation, isolated from any closure or cascade effect.

3.2. *Finite total strain, and the necessity of the rapid term*

Beyond onset the turbulence becomes anisotropic and the rapid term (2.13) ceases to be a function of the Reynolds stress alone, so the verification becomes a test of the spectral closure rather than of the construction. We integrate the model moment equation along each canonical strain with two closures: the baseline isotropic closure (2.14) (equivalently,

	$\mathrm{d}R_{11}/\mathrm{d}t\vert_0$	$\mathrm{d}R_{22}/\mathrm{d}t\vert_0$	$\mathrm{d}R_{33}/\mathrm{d}t\vert_0$
Plane strain (3.1)	$-\frac{8}{15}k_t a$	$+\frac{8}{15}k_t a$	0
Axisymmetric (3.2)	$+\frac{4}{15}k_t \gamma$	$+\frac{4}{15}k_t \gamma$	$-\frac{8}{15}k_t \gamma$

Table 1. Exact onset rates of the component energies from (3.3) for initially isotropic turbulence. The component compressed by the strain is amplified; the stretched component is suppressed.

isotropisation of production with $C_2 = 3/5$; IP) and the quasi-isotropic model of Launder *et al.* (1975) (LRR),

$$\Pi^{(\mathrm{r})}_{ij} = C_2\, k_t S_{ij} + C_3\, k_t \left(b_{ik}S_{jk} + b_{jk}S_{ik} - \tfrac{2}{3} b_{mn}S_{mn}\delta_{ij}\right) + C_4\, k_t \left(b_{ik}W_{jk} + b_{jk}W_{ik}\right), \quad (3.4)$$

with anisotropy $b_{ij} = R_{ij}/2k_t - \frac{1}{3}\delta_{ij}$, mean rotation $W_{ij} = \frac{1}{2}(A_{ij} - A_{ji})$ (zero for the irrotational strains considered), and $C_2 = 4/5$, $C_3 = 7/4$, $C_4 = 131/100$; the value $C_2 = 4/5$ is fixed by requiring that (3.4) reduce at isotropy to the exact $\frac{4}{5}k_t S_{ij}$ of (2.15), so LRR shares the constant-free onset slope. The benchmark is the exact rapid-distortion solution, obtained by integrating the spectral equations (2.8) over a Gauss–Legendre ensemble of initial wavevector directions on the unit sphere: for initially isotropic turbulence the modal dynamics depend only on direction, so the Reynolds stress is the spherical average of the evolved modal covariance. This integration is independent of the moment models and reproduces the onset slope (3.3) to within 10^{-16}, verifying the rapid-distortion machinery itself.

Figure 1 reports the component-energy fractions against total strain. Three observations follow. First, at onset all curves for each component are tangent—the constant-free identity (3.3) is satisfied—and through $e \lesssim 1$ the closures are nearly indistinguishable from the exact solution. Second, as anisotropy accumulates the closures depart, LRR uniformly less than IP: for axisymmetric strain at $e = 4$ the lateral fraction is 0.498 (exact), 0.468 (LRR), 0.450 (IP), and the axial fraction 0.003, 0.063, 0.100. This gap is the spectral-closure error in isolation—the eddy events are inactive in the rapid limit—and it bounds the accuracy attainable from any single rapid operator. Third, and most important to state plainly, for plane strain both closures miss a *qualitative* feature: in the exact solution the neutral spanwise component grows monotonically, overtakes the upwash near $e \approx 3.3$, and is the largest component by $e = 4$ ($\overline{u_3^2}/2k_t = 0.51$ against 0.47), whereas both closures predict it to decay. This redistribution into the neutral direction is governed by the orientation distribution of the spectral tensor and cannot be recovered from single-point quantities; it is the moment-level signature of precisely the transverse-wavenumber information a single line does not carry. Section 5.6 returns to it with the strained-box DNS in hand and localises the deficiency in the closure's b-dependent term.

The upwash component, whose spectrum is the input to the leading-edge response, makes the case for the rapid term quantitative (figure 2). Production acting alone drives the upwash fraction from 1/3 to 0.98 by $e = 4$—almost the entire kinetic energy forced into one component, an essentially non-realisable state—and overstates the initial amplification rate by the factor $\frac{5}{2}$. The rapid term removes three fifths of this at onset and keeps the fraction physical: the exact solution peaks at 0.49 near $e \approx 2.4$ and then relaxes as energy passes to the spanwise component. Both closures reproduce the amplification itself—the

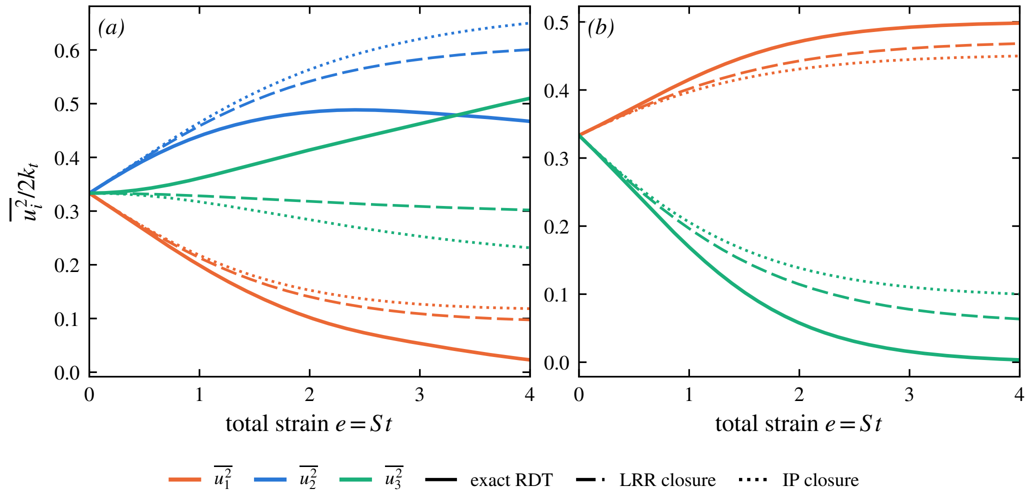

Figure 1. Component energy fractions against total strain for initially isotropic turbulence under (*a*) plane strain and (*b*) axisymmetric strain, normalised to $S = 1$: exact rapid-distortion theory (wavevector-ensemble integration of (2.8)) against the model moment equation with the LRR closure (3.4) and the IP closure (2.14). For axisymmetric strain the two lateral components coincide and only $\overline{u_1^2} = \overline{u_2^2}$ (lateral) and $\overline{u_3^2}$ (axial) are shown. All closures share the exact onset slope; LRR tracks the exact solution more closely at finite strain.

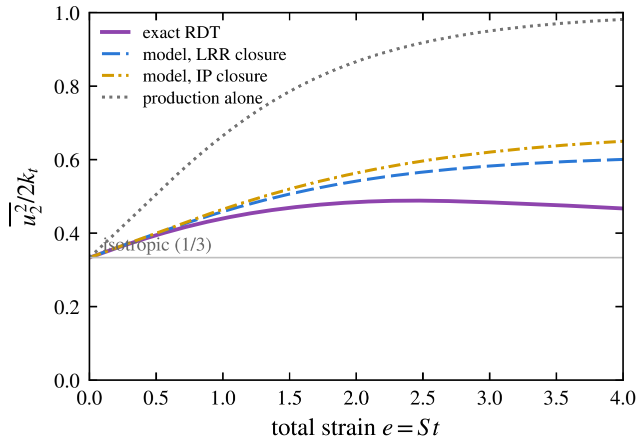

Figure 2. Upwash energy fraction under plane strain for initially isotropic turbulence: exact rapid-distortion theory, the model with the LRR and IP closures, and production alone ($B_{ij} = 0$). Production without the rapid term overstates the onset amplification rate by $\frac{5}{2}$ and forces almost all the energy into the upwash component; the rapid term restores the physical, bounded amplification.

effect that the frozen, isotropic input of classical leading-edge-noise prediction omits—while overpredicting it at large strain (LRR 0.60, IP 0.65 at $e = 4$, against the exact 0.47). The rapid pressure–strain is therefore not a refinement but a necessity, and the accuracy of the method for a given configuration is controlled by the total strain accumulated along the stagnation streamline.

3.3. *Verification of the implementation*

The preceding checks verify the *formulation*. They do not test the *implementation*, in which the operator is not prescribed but reconstructed at every substep: the line Reynolds

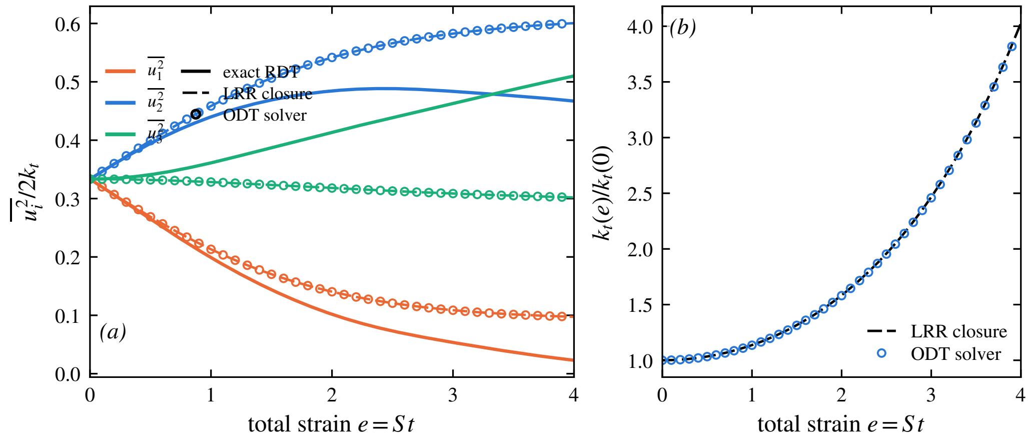

Figure 3. Verification of the strain-coupled implementation under plane strain, eddy events suppressed and zero molecular diffusion. (*a*) Component energy fractions; (*b*) kinetic-energy amplification $k_t(e)/k_t(0)$. Symbols: solver, length-weighted line averages. Dashed: the LRR closure integrated as a moment equation. Solid: exact rapid-distortion theory. The solver reproduces the closure to 10^{-4} in the fractions and 0.3% in the energy; the offset from the exact solution is the closure error of § 3.2, as it must be.

stress is formed from the resolved field, the closure supplies $\Pi_{ij}^{(\mathrm{r})}$, the Lyapunov equation (2.16) is solved for B_{ij}, and $\mathcal{A}_{ij} = -A_{ij} + B_{ij}$ is applied pointwise. To isolate this chain the solver is run in the rapid-distortion limit—plane strain, eddy events suppressed, zero molecular diffusion—from a statistically isotropic resolved initial field, and the component energies are evaluated as length-weighted line averages. With the events inactive the line's component-energy evolution must coincide with the closed moment equation across the whole strain range, not merely at onset.

Figure 3 shows that it does: the solver output lies on the LRR trajectory over $0 \le e \le 4$ to within 10^{-4} in each component fraction, so the per-substep stress reconstruction, Lyapunov solve and operator assembly are correct at every state of anisotropy. The comparison must be read correctly: the solver reproduces the *closure*, and it should—with the events suppressed the line carries no information beyond the single-point statistics the closure itself uses. The offset from the exact solution is the single-point closure error of § 3.2, not an implementation error; agreement with the exact solution here would indicate a fault. The fractions test only the anisotropy; the kinetic energy, which the traceless rapid term cannot change directly and which grows through the production integral $\mathrm{d}\ln k_t/\mathrm{d}e = f_{22} - f_{11}$, provides the independent magnitude check, and the solver reproduces the LRR amplification to within 0.3%. Reaching that figure required a resolved spectral initialisation (a white-noise start places energy at the grid scale, where mesh-adaption interpolation damps it) and a strain-resolved timestep bound in the inviscid limit; both effects act nearly isotropically, so the energy panel is what exposes them.

3.4. *Moment-level validation against the strained-turbulence DNS of Lee & Reynolds*

The DNS of Lee & Reynolds (1985) is retained as a benchmark for one specific reason: it was deliberately run at a turbulence Reynolds number low enough ($q^4/\nu\varepsilon \lesssim 100$) to be fully resolvable on a single line, and it reports the Reynolds-stress anisotropy against total strain for several irrotational strain modes alongside the analytic rapid-distortion prediction—a graded, moment-level reference that the spectral benchmarks of § 4 do not provide. The comparison quantity is the anisotropy b_{ij} against the deformation ratio $c = \exp\int S\,\mathrm{d}t$ (so $\ln c$ is the total strain e of the present notation), computed from the single-point component

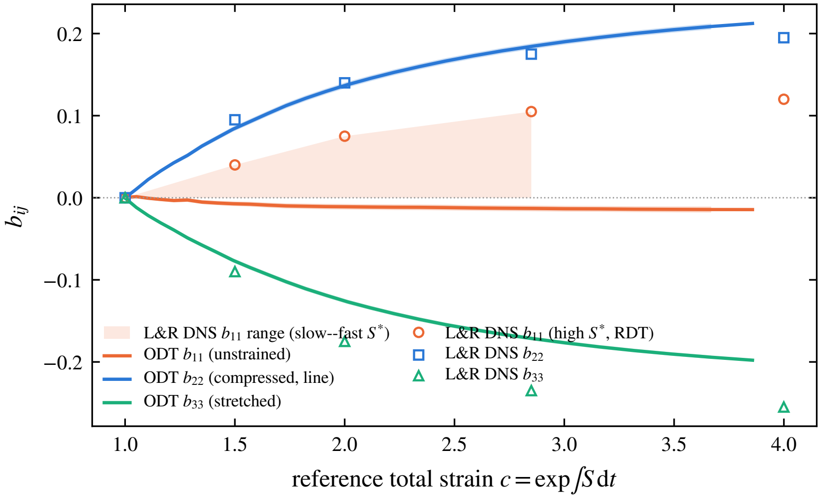

Figure 4. Reynolds-stress anisotropy under plane strain: present model (lines; ensemble mean over 1000 realisations, standard-error band) against the DNS of Lee & Reynolds (1985) (symbols, digitised from their figure 5.4). The strained components b_{22} and b_{33} are reproduced quantitatively; the unstrained b_{11} is strain-rate-dependent in the DNS (shaded band), with the model's near-zero value at the slow-strain edge.

variances along the line, ensemble-averaged over 1000 realisations; no spectral transform is involved. Two of their strain modes are run: plane strain and axisymmetric contraction, the line along a compressed direction in each case, to the total strains of the DNS ($c = 4$ and 3.32); the measured line compression agrees with $\exp(A_{22}t_f)$ to better than 0.3%, confirming the kinematic fidelity of the dilatation. One parameter distinction matters for interpretation: Lee & Reynolds characterise each run by $S^* = Sq^2/\varepsilon = 2\chi$, and find the axisymmetric anisotropy essentially independent of S^*—a clean, parameter-free target—whereas in plane strain the unstrained component b_{11} is strongly S^*-dependent, spanning $b_{11} \approx 0$ (slow) to +0.12 (rapid) in the DNS.

Under plane strain (figure 4) the model reproduces the two strained components quantitatively across the whole range: $b_{22} = +0.21$ at $c = 3.86$ against $\approx +0.20$ in the DNS, and $b_{33} = -0.20$ against ≈ -0.26, capturing the trend and the bulk of the magnitude. These are exactly the components the DNS identifies as strain-rate-independent, so they constitute the robust part of the comparison. Under axisymmetric contraction (figure 5) the model reproduces the sign structure and the monotonic approach to a strained state while under-predicting the anisotropy magnitude by roughly 25% ($b_{11} = -0.21$ against ≈ -0.29 at $c = 3.16$); the dominant contribution is the spectral-closure gap isolated in § 3.2, common to the whole second-moment family, with a small additional artefact specific to the line: lying along one of the two equivalent transverse directions, it breaks the $x_2 \leftrightarrow x_3$ symmetry and returns a residual split $b_{22} = +0.102$ against $b_{33} = +0.113$ at $c = 3.16$, which the symmetry-respecting mean $\frac{1}{2}(b_{22} + b_{33})$ removes.

The one component the model does not reproduce, the plane-strain unstrained anisotropy b_{11}, is not a deficiency specific to the model, and figure 6 locates it exactly. Exact linear rapid distortion, evaluated by the wavevector-ensemble integration of § 3.2, gives b_{11} *rising* to +0.114 at $c = 4$: although $A_{1k} = 0$, the rapid pressure–strain feeds the neutral direction—the same orientation-carried redistribution as the spanwise growth in § 3.2. (A tempting kinematic argument, that $\langle u_1^2 \rangle$ is conserved because the component is unstrained, holds for production but not for the rapid term, and gives the wrong sign; an earlier version of this comparison used it.) The fast-strain DNS therefore *tracks exact RDT*, as Lee &

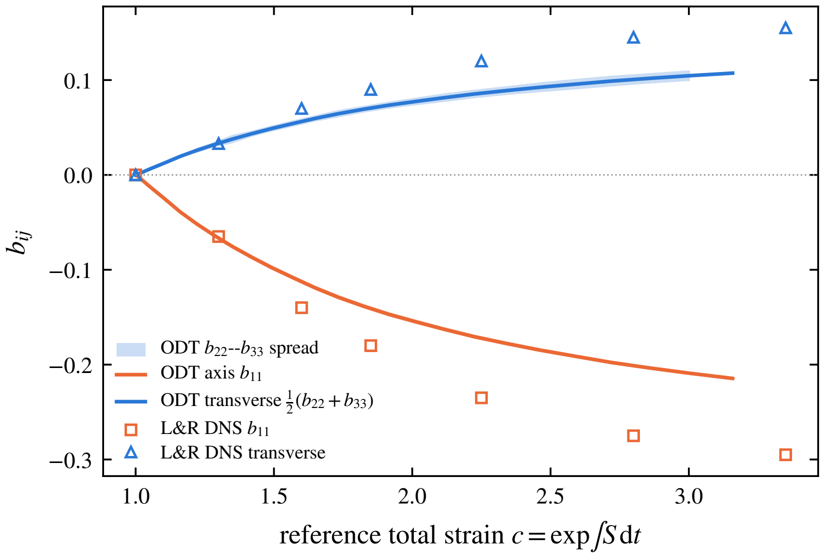

Figure 5. Anisotropy under axisymmetric contraction: present model (lines; 1000 realisations) against Lee & Reynolds (1985) (symbols, digitised from their figure 5.8). The shaded spread is the residual $b_{22} \neq b_{33}$ split from the line's breaking of the transverse symmetry.

Reynolds themselves observed, while both single-point closures give $b_{11} \approx 0$, slightly negative: the closure family misses the neutral-direction feed, which is carried by the orientation distribution that single-point models discard—the same deficiency measured spectrally, and localised in the closure's b-dependent term, in § 5.6. The present model coincides with the closure on which its operator is built, so its near-zero b_{11} is correct for its formulation and matches the DNS at slow strain rate, where the nonlinear scrambling erases the neutral-direction feed; at fast strain it inherits the closure family's known limitation rather than any line-specific artefact. The components that govern the upwash spectrum fed to the leading-edge response are precisely those the model reproduces quantitatively. Finally, Lee & Reynolds's own transport-budget analysis finds that in plane strain the intercomponent transfer is carried almost entirely by the rapid term—independent support for the rapid/slow decomposition on which § 5 rests.

4. The linear reference and the distorted spectrum

4.0.1. *Compression without cascade: the model's kinematic reference*

With the eddy events suppressed and zero molecular viscosity, the mean strain is the only process acting on the resolved field, and its effect on the line is purely kinematic. The line length contracts as $\mathcal{L}(e) = \mathcal{L}_0\, e^{A_{22}e}$, so every resolved wavenumber is rescaled as $k(e) = k(0)\, e^{-A_{22}e}$—the wavevector kinematics it shares with rapid-distortion theory (Batchelor & Proudman 1954; Hunt & Carruthers 1990)—while the continuous redistribution operator of § 2.4 acts uniformly on every resolved wavenumber. The combination transfers no energy between scales: a rigid translation of the line spectrum toward higher wavenumber. We emphasise at the outset that this rigid translation is a property of the strain-coupled *line*—uniform dilatation composed with a wavenumber-uniform operator—and not of linear distortion theory itself; the correct linear reference for the projected spectrum is constructed in § 4.0.2 below. As a check on the implementation, however, the rigid translation is an exact, parameter-free target.

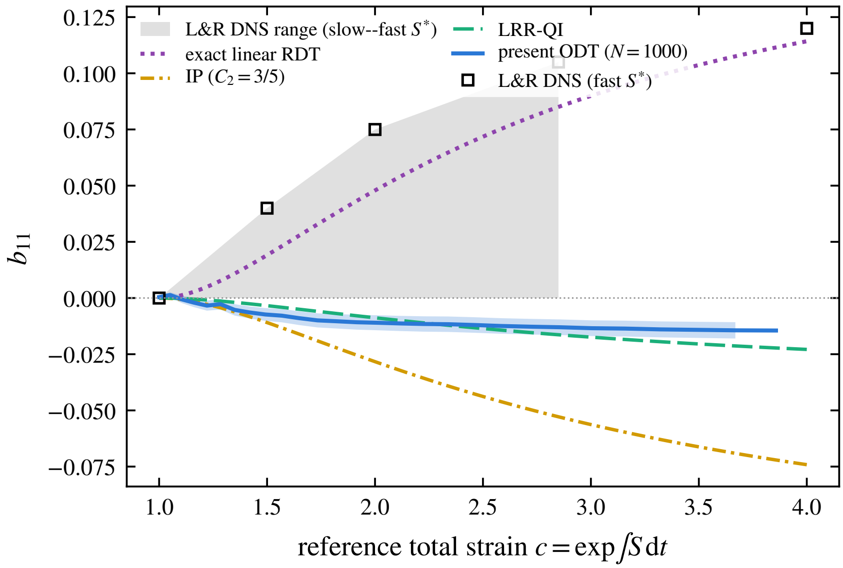

Figure 6. The unstrained-direction anisotropy b_{11} under plane strain: present model (solid, 1000 realisations, standard-error band) against exact linear rapid distortion (dotted, wavevector-ensemble integration), the IP and LRR-QI closures, and the DNS of Lee & Reynolds (1985) (symbols and shaded S^* range). The fast-strain DNS tracks exact RDT, whose rapid pressure–strain feeds the neutral direction; the single-point closures—and the model built on them—miss this orientation-carried redistribution, the deficiency measured and localised in § 5.6.

Figure 8 confirms it. The component spectra translate toward higher k without change of shape—the peak neither broadens nor develops an inertial range—and the spectral centroid follows $\mathrm{e}^{-A_{22}e}$, reaching 7.03 at $e = 3.9$ against the exact 7.03, with the three components migrating identically; the domain length follows $\mathrm{e}^{A_{22}e}$ to four digits. The single-point moments are unchanged from the no-dilatation case, since a uniform rescaling of position leaves the length-weighted velocity statistics invariant. Compression therefore acts entirely on the spectrum and reproduces the model's kinematic prediction exactly. It is the kinematic reference (the dashed line in all centroid figures below) against which the full model is measured.

4.0.2. *The linear reference for the projected spectrum: exact RDT*

The rigid translation above must not be mistaken for the prediction of linear rapid-distortion theory for the quantity the line actually carries. In the full linear theory the amplitude of each Fourier mode evolves at a rate that depends on the orientation of its three-dimensional wavevector (equation 2.8); the one-dimensional component spectra are projections—integrals over the transverse wavenumber plane—and there is no reason for a projection of orientation-dependent amplification to translate rigidly. Whether it does is a question of calculation, and the answer is that it does not. Using the closed-form Cauchy solution of the linear problem applied to isotropic initial fields and projected onto a line (the exact-RDT companion calculations described in § 4.1), we define the shape ratio $R_{nn}(\kappa_2)$ as the projected component spectrum at strain e, normalised by its integral, divided by the rigidly translated initial spectrum normalised likewise; $R_{nn} \equiv 1$ at every κ_2 if and only if the translation is rigid. At $e = 1$ under plane strain the upwash ratio R_{22} falls monotonically from 1.60 at half the initial spectral centroid to 0.38 at twelve times it: exact linear theory redistributes the projected upwash spectrum toward low wavenumber

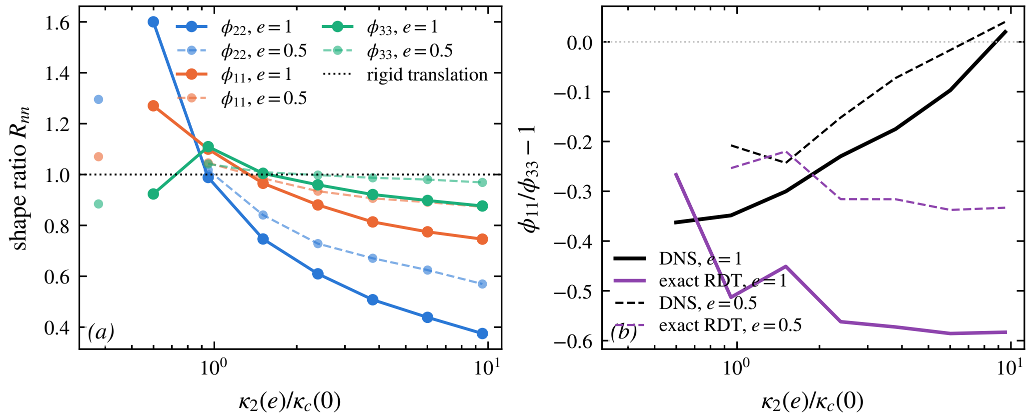

Figure 7. Exact linear theory does not translate the projected spectra rigidly. Left: shape ratio $R_{nn}(\kappa_2)$ of the exactly distorted, line-projected component spectra to a rigid translation, at $e = 1$ under plane strain (closed-form Cauchy solution, isotropic initial fields, four realisations). Right: the transverse splitting $\phi_{11}/\phi_{33}-1$ versus wavenumber in exact linear theory and in the strained-box DNS of § 4.1.

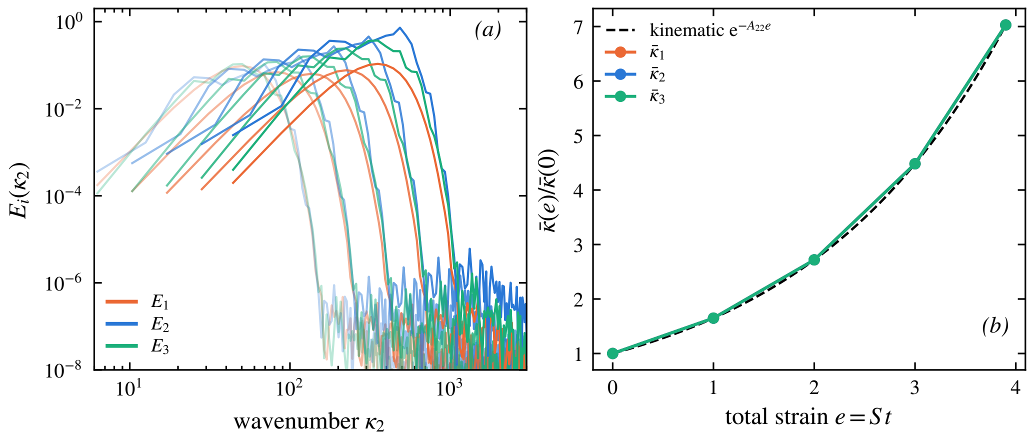

Figure 8. Compression without cascade (eddy events suppressed, inviscid). (*a*) Component line spectra $E_i(\kappa_2)$ at total strains $e = 0, 1, 2, 3, 3.9$ (lighter to darker); the spectra translate to higher wavenumber without changing shape. (*b*) Spectral-centroid migration $\bar{\kappa}(e)/\bar{\kappa}(0)$ for each component (symbols) against the kinematic prediction $\mathrm{e}^{-A_{22}e}$ (dashed). The centroid follows the prediction and the three components coincide, confirming purely geometric compression of the line.

by a factor of four across the resolved range, because modes aligned with the line ($\kappa \to \kappa_2$) are amplified least. The transverse components are affected more weakly (R_{11} from 1.27 to 0.75; R_{33} within ten per cent of unity), so the projected anisotropy of linear theory is itself scale-dependent. Figure 7 shows the ratios.

Two consequences run through the remainder of the paper. First, the comparison of the full model against the rigid translation (§ **??**) measures the departure of the model from its own kinematics; the departure of the *flow* from linear theory must be measured against the exact projected reference, and the two differ already at the linear level. Second, the model's rigid spectral kinematics is itself a structural approximation—linear theory deepens the projected anisotropy with wavenumber, which a wavenumber-uniform operator cannot represent—and § 4.1.1 quantifies the resulting deficiency against both the exact reference and DNS.

$\Delta b_{22}(\kappa_2)$, band κ_2/κ_c:	0.5–0.75	0.75–1.2	1.2–1.9	1.9–3	3–4.8	4.8–7.6	7.6–12
exact projected RDT	+0.146	+0.008	−0.020	−0.041	−0.050	−0.054	−0.056
DNS	+0.032	−0.033	−0.029	−0.032	−0.036	−0.040	−0.046
strain-coupled ODT	+0.100	+0.101	+0.114	+0.111	+0.127	+0.099	+0.147

Table 2. Strain-induced upwash anisotropy per wavenumber band at $e = 1$, $Sk_t/\varepsilon = 0.8$, each system referenced to its own unstrained state. Bands below $0.75\,\kappa_c$ rest on a few box modes and are indicative only.

[M-spectrum: transplant sec:spec-bvc + compressed CMK here.]

4.1. *Validation against a strained-box DNS with an exact-RDT companion*

The comparisons above validate the model at the Reynolds-stress level. The referent quantity for the leading-edge application, however, is spectral, and the objection that the emergent-spectrum results of § 4 compare the model only with itself must be met with an independent spectral benchmark under the same strain. To that end we performed direct numerical simulations of plane-strained homogeneous turbulence in a deforming-frame (Rogallo-type) pseudo-spectral formulation: a 128^3 box, decaying isotropic precursor, then plane strain $A = S\,\mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ applied at strain-rate parameters $Sk_t/\varepsilon \in \{0.8, 16\}$ at onset, carried to total strain $e = 1$, four independent realisations ($\kappa_{\max}\eta \approx 0.76$ at $e = 1$: a pilot resolution, so small-scale statements below are qualitative). Alongside every DNS realisation runs an *exact-RDT companion*: the same initial field evolved under the closed-form Cauchy solution of the linearised equations and projected onto lines identically to the DNS and to ODT. The companion is what § 4.0.2 used to construct the linear reference; here it completes a three-way comparison in which model, linear theory and DNS are reduced to the same one-dimensional observables. On the ODT side the ensembles use an unstrained precursor: the line is evolved as ordinary ODT until the initialisation transient has relaxed and the eddy population is statistically active, and the strain is switched on only then, so that the strained evolution starts from a physically conditioned state rather than from the synthetic initial condition. All ODT statistics in this subsection are medians over 1024 realisations per case with bootstrap confidence intervals; median statistics are used because the strained band energies are strongly intermittent across realisations, and ensemble means of spectral ratios are dominated by rare events at any practical sample size.

4.1.1. *Three-way spectral anisotropy: ODT, exact RDT, DNS*

The comparison quantity is the strain-induced change of the spectral component anisotropy, $\Delta b_{nn}(\kappa_2) = \phi_{nn}/\sum_m \phi_{mm} - \frac{1}{3}$ at strain e, minus the same quantity evaluated in the system's own unstrained state with each mode mapped to its strained wavenumber. Referencing each system to its own $e = 0$ state cancels viscous decay, which is component-blind, and the longitudinal–transverse structure of the isotropic line spectra, which differs between a three-dimensional box and the component-symmetric ODT line. Figure 9 and table 2 give the result at $e = 1$, $Sk_t/\varepsilon = 0.8$, on common bands of $\kappa_2(e)/\kappa_c(0)$ with κ_c the initial spectral centroid.

The structure of the table is the finding. Exact linear theory concentrates the strain-induced upwash anisotropy at the energy-containing scales and *removes* it from the small scales, because modes aligned with the line are amplified least; the DNS shows the same shape, weakened at the large scales by the nonlinear return. The strain-coupled ODT distributes its moment-level anisotropy essentially uniformly over wavenumber—the

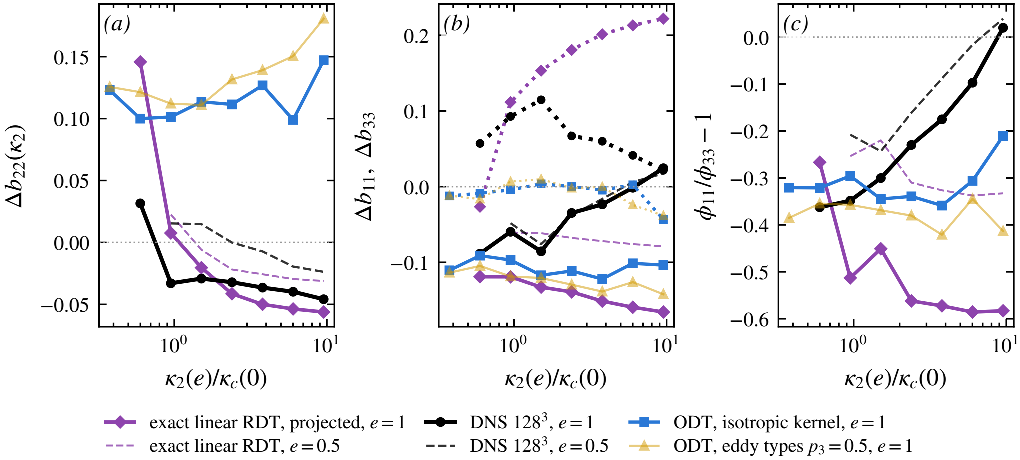

Figure 9. Three-way comparison at $Sk_t/\varepsilon = 0.8$: strain-induced anisotropy of the line spectra relative to each system's own unstrained state; strain-coupled ODT (1024 realisations) against exact projected RDT and DNS (128^3, four realisations); $e = 1$ solid, $e = 0.5$ dashed.

signature, anticipated in § 4.0.2, of a wavenumber-uniform redistribution operator acting on a uniformly dilated line. The integrated anisotropy is nevertheless correct (the model tracks the closure at the moment level, § 3.4), so the deficiency is one of allocation across scales, not of amount. For the leading-edge application the operative consequence is that the high-wavenumber anisotropy of the upwash spectrum, which the Amiet response weights, has the wrong sign in the present model; the closure element that must carry the correction—a wavenumber-dependent rapid operator in place of the uniform one—is exactly the spectral kernel formulation of § 5, whose exercise is thereby motivated by a measured deficit rather than by completeness.

4.1.2. *Distance from linear theory as a function of strain rapidity*

The three-way comparison resolves the anisotropy by wavenumber at one strain-rate parameter. A complementary scalar measure tracks the *whole* spectral shape across strain and strain rapidity: at each total strain the median ODT component line spectra are fitted, jointly in the upwash and transverse components, by the exactly distorted spectrum of linear theory—the closed-form Cauchy distortion of an isotropic von Kármán field, projected onto the line, with only the pre-distortion amplitude and energy wavenumber free. The root-mean-square log-residual of that two-parameter fit is a scale-resolved distance between the model's strained spectra and exact linear theory. Figure 10 shows it for precursor-conditioned ensembles at $Sk_t/\varepsilon \approx 0.4$ and ≈ 8 at onset (1024 realisations each, identical precursor state: the two fits coincide at $e = 0$ at thirteen per cent, the residual shape mismatch of a relaxed ODT line against the von Kármán form).

The distance grows with strain in both cases, but *faster at rapid strain*: to approximately 22 per cent at $e = 2$ for $Sk_t/\varepsilon \approx 0.4$ against approximately 24 per cent, saturating early, at $Sk_t/\varepsilon \approx 8$, with the separation established already by $e = 0.5$. If the model's rapid response were spectrally faithful, the rapid curve is the one that should approach zero: eddy events are subdominant there by construction. That the deviation is largest precisely where the eddies matter least localises the deficiency in the rapid kinematics themselves—the wavenumber-uniform operator and the rigid dilatation—and not in the eddy or relaxation statistics, in quantitative agreement with the three-way comparison. This measured validity

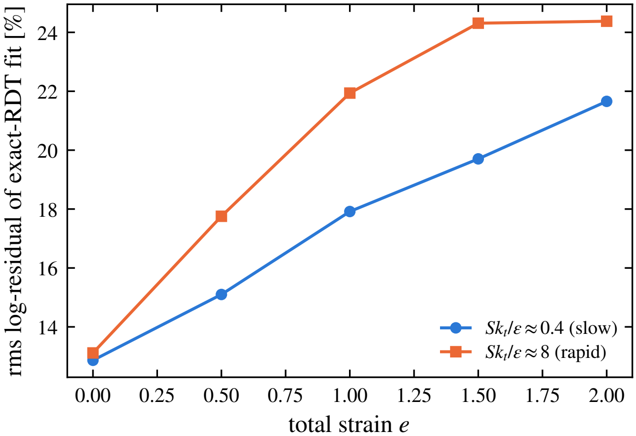

Figure 10. Scale-resolved distance between the strained ODT line spectra and exact linear theory: rms log-residual of the two-parameter exact-RDT-distorted von Kármán fit, against total strain, for slow and rapid onset strain-rate parameters (medians over 1024 realisations; identical precursor state at $e = 0$).

envelope accompanies the model into any application: predictions inherit a spectral-shape uncertainty of order the plotted distance at the operating point's strain and rapidity.

5. Closure of the rapid pressure–strain term on a single line

The results of § 4 establish what the strain-coupled model produces and where it stands against exact linear theory and DNS. The rapid pressure–strain redistribution that drives that evolution was evaluated throughout by the algebraic moment closure of § 2 (Launder *et al.* 1975), applied component by component on the resolved line. That closure is a modelling input, and the redistribution it represents is, in its exact form, a *nonlocal, three-dimensional* object: the rapid pressure at a point is a volume integral over the entire fluctuating field, and the pressure–strain correlation is an integral over the full three-dimensional spectrum. What a single one-dimensional line of turbulence can say about such an object is the analytical question of this paper, and this section answers it in a form that is deliberately more modest, and more defensible, than a closure claim: we state exactly what obstructs a single-line representation, we derive sharp bounds on the rapid term from line data alone, we *measure* rather than assume the axisymmetry hypothesis under which the term collapses, and we identify—by measurement—which physical content the moment closure discards.

The section proceeds as follows. The exact term and the obstruction are stated first (§ 5.1–§ 5.2); the axisymmetric collapse follows, together with the statement of what it does *not* deliver (§ 5.3); the bounds that line data do deliver, with their falsifiers, come next (§ 5.4); the axisymmetry assumption is then measured directly in the strained-box DNS, including a control experiment that isolates its failure mode (§ 5.5); the component-ordering deficiency raised by the exact solution is measured and localised (§ 5.6); the reduction to the moment closure and the disjointness from the eddy redistribution close the argument (§ 5.7–§ 5.8).

5.1. *The exact rapid pressure–strain term*

The starting point is the Poisson equation (2.7) for the fluctuating pressure, whose source splits into a part linear in the fluctuations and the mean velocity gradient (the *rapid* part) and a part quadratic in the fluctuations (the *slow* part) (Pope 2000). The rapid part is linear in $\boldsymbol{u}'$ and proportional to the mean gradient, which for the present homogeneous plane strain is the constant tensor A_{kl}. Writing p'_{r} for the rapid pressure and inverting the Laplacian with the free-space Green's function $G(\boldsymbol{x}) = -1/4\pi|\boldsymbol{x}|$,

$$p'_{\mathrm{r}}(\boldsymbol{x}) = \frac{\rho}{2\pi} A_{kl} \int \frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|} \frac{\partial u'_l}{\partial x'_k}(\boldsymbol{x}')\, \mathrm{d}\boldsymbol{x}'. \tag{5.1}$$

The rapid pressure–strain correlation that redistributes energy among the Reynolds-stress components is $\Pi^{(\mathrm{r})}_{ij} = \overline{(p'_{\mathrm{r}}/\rho)\,(\partial u'_i/\partial x_j + \partial u'_j/\partial x_i)}$. Substituting (5.1) and using homogeneity to pass to the velocity spectrum tensor $\Phi_{mn}(\boldsymbol{\kappa})$ (the Fourier transform of $\overline{u'_m(\boldsymbol{x})\, u'_n(\boldsymbol{x}+\boldsymbol{r})}$), one obtains the standard exact result (Batchelor & Proudman 1954; Crow 1968; Hunt & Carruthers 1990)

$$\Pi^{(\mathrm{r})}_{ij} = A_{kl}\, M_{ijkl}, \qquad M_{ijkl} = \int_{\mathbb{R}^3} \left[\frac{\kappa_j \kappa_k}{\kappa^2}\, \Phi_{il}(\boldsymbol{\kappa}) + \frac{\kappa_i \kappa_k}{\kappa^2}\, \Phi_{jl}(\boldsymbol{\kappa}) \right] \mathrm{d}\boldsymbol{\kappa}, \tag{5.2}$$

where $\kappa = |\boldsymbol{\kappa}|$. Equations (2.7)–(5.2) are exact; no modelling has entered. The tensor M_{ijkl} is a linear functional of the *full three-dimensional* spectrum $\Phi_{mn}(\boldsymbol{\kappa})$ through the nonlocal projection operator $\kappa_a\kappa_b/\kappa^2$, which is the wavenumber-space image of the inverse Laplacian in (5.1). This nonlocality and three-dimensionality are the obstruction that the single-line model must confront.

5.2. *The obstruction: what an ODT line resolves*

ODT evolves a velocity vector $\boldsymbol{u}'(y)$ on a single line aligned with the resolved direction, here taken along the contraction axis of the strain, $y \equiv x_2$. From it the model has access to the one-dimensional spectra

$$\phi_{mn}(\kappa_2) = \int_{\mathbb{R}^2} \Phi_{mn}(\boldsymbol{\kappa})\, \mathrm{d}\kappa_1\, \mathrm{d}\kappa_3, \tag{5.3}$$

the one-dimensional spectra obtained by integrating Φ_{mn} over the two unresolved wavenumber components. The integrand of (5.2), by contrast, carries the *direction-resolved* weight $\kappa_a\kappa_b/\kappa^2$, which cannot be reconstructed from $\phi_{mn}(\kappa_2)$ alone: integrating out κ_1, κ_3 as in (5.3) destroys exactly the angular information that the projection operator acts on. The reduction $M_{ijkl} \rightarrow$(functional of ϕ_{mn}) is therefore not available without an assumption about how the energy is distributed over the unresolved wavevector directions. This is the crux of the present question, and we make the required assumption explicit rather than implicit.

The obstruction just stated does not disappear under any symmetry assumption short of full isotropy. As § 5.3 shows, axisymmetry about the line reduces the unknown spectrum to two scalar functions of $(\kappa_2, \kappa_\perp)$—but the line data are two weighted $\kappa_\perp$-integrals of those functions per resolved wavenumber, so the map from what the line carries to what the rapid term requires retains an infinite-dimensional null space. Stating this null space exactly, and extracting what survives it, is the correct formulation of the single-line question.

5.3. *The closure assumption and the collapsed form*

The spectrum tensor of the incoming turbulence, before the rapid distortion acts, is axisymmetric about the resolved direction $\boldsymbol{e}_2$: $\Phi_{mn}(\boldsymbol{\kappa})$ depends on $\boldsymbol{\kappa}$ only through κ_2 and $\kappa_\perp = (\kappa_1^2 + \kappa_3^2)^{1/2}$, and its polarization is invariant under rotation about $\boldsymbol{e}_2$.

This is a statement about the *turbulence*, not about the strain. It is justified for the present configuration because the inflow is grid turbulence, which is closely isotropic at the grid and is subsequently contracted along a single axis as it approaches the stagnation plane; isotropy is the special case of A1, and uniaxial contraction preserves axisymmetry about the contraction axis. The plane strain $A_{kl} = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ acting at the leading edge is itself *not* axisymmetric, so A1 is imposed on the state on which the rapid operator acts, consistent with the rapid-distortion ordering in which $\Pi^{(\mathrm{r})}$ is evaluated on the current (here, axisymmetric to leading order) field. The error incurred when the accumulated distortion has driven the spectrum away from axisymmetry is quantified in § 5.5.

Under A1 the angular integral in (5.2) can be carried out. Writing $\boldsymbol{\kappa} = (\kappa_\perp \cos\theta,\ \kappa_2,\ \kappa_\perp \sin\theta)$ and using that, by A1, Φ_{mn} is independent of θ, the azimuthal averages of the projection weights reduce to

$$\left\langle \frac{\kappa_1^2}{\kappa^2} \right\rangle_\theta = \left\langle \frac{\kappa_3^2}{\kappa^2} \right\rangle_\theta = \frac{1}{2}\,\frac{\kappa_\perp^2}{\kappa^2}, \qquad \left\langle \frac{\kappa_2^2}{\kappa^2} \right\rangle_\theta = \frac{\kappa_2^2}{\kappa^2}, \qquad \left\langle \frac{\kappa_1\kappa_3}{\kappa^2} \right\rangle_\theta = 0, \tag{5.4}$$

with $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and all weights mixing a resolved and an unresolved index (e.g. $\kappa_1\kappa_2/\kappa^2$) averaging to zero. Substituting (5.4) into (5.2) removes the azimuthal dependence and leaves a two-dimensional integral over $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. Defining the diagonal one-dimensional spectra along the line,

$$E_n^{\parallel}(\kappa_2) = \int_0^\infty \Phi_{nn}(\kappa_2, \kappa_\perp)\, 2\pi\kappa_\perp\, \mathrm{d}\kappa_\perp \quad \text{(no sum)}, \tag{5.5}$$

the rapid pressure–strain term reduces to a two-dimensional integral over the half-plane $(\kappa_2, \kappa_\perp)$ of the axisymmetric spectrum. The axisymmetric, solenoidal spectrum is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950); we denote them $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$, where a is the isotropic-like scalar (the sole survivor when the turbulence is isotropic) and c measures the axisymmetric departure from isotropy. Carrying out the azimuthal average (5.4) and specialising to the plane strain $A = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$, the diagonal rapid pressure–strain components are

$$\Pi_{nn}^{(\mathrm{r})} = \int_0^\infty\!\!\int_0^\infty g_n(x)\left[\, a(\kappa_2,\kappa_\perp),\ c(\kappa_2,\kappa_\perp) \,\right] 2\pi\kappa_\perp\, \mathrm{d}\kappa_\perp\, \mathrm{d}\kappa_2 \quad \text{(no sum on } n\text{)}, \tag{5.6}$$

with $x \equiv \kappa_2^2/\kappa^2$, $\kappa^2 = \kappa_2^2 + \kappa_\perp^2$, and the kernels, linear in the two spectral scalars,

$$g_1(x) = a\left(\tfrac{1}{8} + \tfrac{3}{4}x - \tfrac{7}{8}x^2\right) + c\,\tfrac{7}{8}\,x\,(1-x)^2, \tag{5.7}$$

$$g_2(x) = -\tfrac{3}{2}\,x\left[\, a\,(1-x) + c\,(1-x)^2 \,\right], \tag{5.8}$$

$$g_3(x) = a\left(-\tfrac{1}{8} + \tfrac{3}{4}x - \tfrac{5}{8}x^2\right) + c\,\tfrac{5}{8}\,x\,(1-x)^2. \tag{5.9}$$

Equations (5.6)–(5.9) constitute the axisymmetric collapse: *under A1, the rapid pressure–strain term is a functional of the axisymmetric spectral scalars a, c on the resolved half-plane and the mean strain*, with no reference to the unresolved azimuthal structure beyond what A1 fixes. The kernels satisfy two exact constraints that verify the reduction. First, energy conservation: pressure–strain redistributes without changing the trace, and indeed $g_1(x)+g_2(x)+g_3(x) = 0$ identically for all x and for both scalars, so $\Pi_{11}^{(\mathrm{r})}+\Pi_{22}^{(\mathrm{r})}+\Pi_{33}^{(\mathrm{r})} = 0$. Second, the isotropic limit: setting $c = 0$ with a a function of κ alone, the integral of g_3 over the half-plane vanishes, so $\Pi_{33}^{(\mathrm{r})} = 0$—as it must, since the plane strain has no component along x_3 and the isotropic rapid term is proportional to S_{ij} (Crow 1968). Both constraints hold identically in the derived kernels, confirming the tensor reduction.

The status of (5.6)–(5.9) must be stated precisely, because it is here that the closure question is decided. Under A1 the rapid term is a linear functional of the two axisymmetric scalars $a(\kappa_2, \kappa_\perp)$ and $c(\kappa_2, \kappa_\perp)$ on the resolved half-plane—a *two-dimensional* representation, not yet a single-line closure, since the line determines only the $\kappa_\perp$-integrals (5.5) of fixed combinations of a and c and not the functions themselves. Equations (5.6)–(5.9) are therefore the *collapsed form* on which any single-line statement must be built; the single-line content itself—what the line data pin down, and how sharply—is established next.

5.4. *What line data determine: sharp bounds on the rapid term*

Under A1 the solenoidal spectrum tensor takes the two-scalar form $\Phi_{mn} = a\,P_{mn} + c\,\xi_m\xi_n$, with P_{mn} the solenoidal projector and ξ_m the projected axis direction; realisability is the pair of pointwise conditions $a \geq 0$ and $a + (1-x)\,c \geq 0$ with $x = \kappa_2^2/\kappa^2$. The line carries $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ and $\phi_{22}(\kappa_2)$: per resolved wavenumber, two weighted $\kappa_\perp$-integrals of the two unknown functions. The rapid term (5.6) is linear in (a, c), so its extreme values over everything consistent with the line data, solenoidality and realisability are obtained from a family of linear programs, one per κ_2, yielding sharp bounds $[\Pi^-_{nn}, \Pi^+_{nn}]$ and a kinematic indeterminacy $\mathcal{I}_n = (\Pi^+_{nn} - \Pi^-_{nn})/2k_t S$ that quantifies exactly how much the null space costs. The forward map from (a, c) to the line spectra and the derivation of the bound kernels are given in Appendix A.

Calibrated on an isotropic von Kármán spectrum (with a verification suite pinning tracelessness, the isotropic angular averages $+\frac{1}{5} : -\frac{1}{5} : 0$, the longitudinal–transverse relation, and the exact $\frac{4}{5}k_t S_{ij}$), the indeterminacy is

constraint level	$\mathcal{I}_{11}$	$\mathcal{I}_{22}$	$\mathcal{I}_{33}$
line data + solenoidality + realisability	0.77	1.28	0.86
+ log-Lipschitz slope cap ($\lambda = 4$)	< 5% narrower		
+ polarisation cap $\lvert c\rvert \leq \gamma a$, $\gamma = 0$	0.34	0.58	0.24

Two features of this table matter. Smoothness assumptions buy almost nothing: the null space is not a roughness artefact. The polarisation cap is the lever—but § 5.5 shows that under strain the effective polarisation reaches order one, so the honest bound in the strained regime is the uncapped one.

The representation also supplies two *free falsifiers* that the line itself reports: under A1, $\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2)$ at every wavenumber (T1) and the velocity cross-spectra vanish (T2). Any measured transverse splitting is a wavenumber-resolved violation of A1; rather than assuming the hypothesis, the line monitors it.

5.5. *The axisymmetry hypothesis, measured*

A1 is exact for the isotropic inflow and is degraded by the plane strain itself. Rather than estimating this error perturbatively, we measure it in the strained-box DNS of § 4.1, where the full spectrum is available: the exact azimuthal decomposition of Φ_{22} yields the true scalars (a, c), the discarded $m = 2$ azimuthal residue (the direct A1-violation measure, with isotropic sampling floor 0.133 in this box), and the effective polarisation $\gamma_{\text{eff}} = |c|/a$.

Three conclusions follow (table 3). First, the A1 error is *not* perturbatively small in the accumulated anisotropy: the residue grows from its floor to 0.19 at $Sk_t/\varepsilon = 0.8$ and to 0.40 at 16 by $e = 1$—larger at rapid strain, because at moderate strain rate the nonlinear dynamics restore axisymmetry as fast as the strain destroys it. An $O(b^2)$ error estimate, which the symmetric structure of the discarded harmonic suggests, is superseded by this measurement. Second, the polarisation cap that narrows the isotropic bounds is illegitimate

	$Sk_t/\varepsilon = 0.8$				$Sk_t/\varepsilon = 16$			
e	b_{22}	γ_{eff}	m=2	$\frac{\phi_{11}}{\phi_{33}}-1$	b_{22}	γ_{eff}	m=2	$\frac{\phi_{11}}{\phi_{33}}-1$
0	0.003	0.10	0.133	+0.06	0.003	0.10	0.133	+0.06
0.5	0.064	0.79	0.166	−0.20	0.070	1.28	0.247	−0.21
1	0.125	1.44	0.189	−0.32	0.124	4.22	0.396	−0.46

Table 3. A1 measured under plane strain (128^3 DNS, four realisations): upwash anisotropy, effective polarisation, azimuthal $m = 2$ residue of Φ_{22} (isotropic floor 0.133), and the on-line falsifier T1.

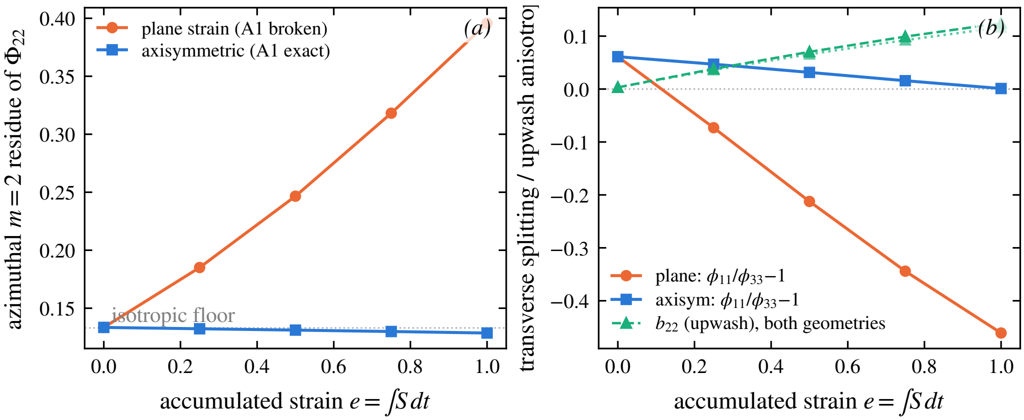

Figure 11. The A1 control experiment at $Sk_t/\varepsilon = 16$ (128^3, four realisations): axisymmetric contraction (A1 exact by symmetry) against plane strain at matched A_{22}. Left: azimuthal $m = 2$ residue of Φ_{22}. Right: transverse splitting and b_{22}.

under strain: γ_{eff} passes unity by $e \simeq 0.5$. Third, correspondingly, the uncapped bound brackets the exact rapid term (evaluated by direct integration in the DNS) up to $e \approx 0.25$–0.5 and fails beyond, and the failure is A1's, not the estimator's.

The decisive evidence for that attribution is a control experiment: the same box under *axisymmetric contraction* with matched A_{22}, for which A1 is exact by symmetry. There the $m = 2$ residue stays at the sampling floor and the transverse splitting at zero at every strain up to $e = 1$ even at $Sk_t/\varepsilon = 16$, while under plane strain both grow (figure 11). The single-line closure is therefore *exact for axisymmetric distortion*; its plane-strain error is a strain-geometry property, now quantified, rather than a defect of the line representation. For the applications this maps cleanly: a rounded, three-dimensional stagnation region imposes an axisymmetric strain (closure exact); a two-dimensional leading edge imposes plane strain (closure carries the measured correction). The acoustically relevant amplification $b_{22}(e)$, finally, is indistinguishable between the two geometries (0.124 against 0.117 at $e = 1$, $Sk_t/\varepsilon = 16$): the quantity the leading-edge response consumes is robust to precisely the geometry that A1 is sensitive to.

5.6. *Component ordering under plane strain*

The exact solution of the rapid problem under plane strain has the neutral spanwise component eventually overtaking the upwash; algebraic closures predict the opposite. The exact rapid term evaluated in the DNS settles the matter: at $e = 1$, $\Pi^{(\mathrm{r})}_{33}/k_t S = 0.14$ at

Sk_t/ε = 0.8 and 0.27 at 16, exceeding $\Pi_{11}^{(\mathrm{r})} = 0.23$ in the rapid case—the overtaking is real and sets in at rapid strain. LRR-QI, whose spanwise component is carried by its b-dependent term, matches at $Sk_t/\varepsilon = 0.8$ ($\Pi_{33}^{(\mathrm{r})} = 0.143\, k_t S$) but delivers half the exact value at 16; IP gives zero identically. The deficiency thus lives in the moment closure's b-term, not in the line model: the ODT moment-level b_{ij} follows the closure to $\sim 2\%$, and its b_{33} stays within 0.006 of zero in every run. In the spectral representation the ordering is carried by the c-scalar through the kernels (5.7)–(5.9); the moment reduction discards it. This answers, by measurement, where the known component-ordering error enters and what it would take to remove it.

5.7. *Reduction to the moment closure*

The closed form (5.6)–(5.9) operates on the resolved spectrum, whereas the distortion results of § 4 were obtained with the algebraic moment closure of § 2.4 (Launder *et al.* 1975). For the two to be consistent, the spectral kernel must reduce to that moment closure in the limit where the moment closure is exact—isotropic turbulence. We show that it does, which both justifies the closure used in the simulations and fixes the prefactor in an explicit convention.

Figure 12 plots the kernels g_1, g_2, g_3 of (5.7)–(5.9) against the angular variable $x = \kappa_2^2/\kappa^2$. Two features are exact and hold for both spectral scalars. The kernels sum to zero at every x,

$$g_1(x) + g_2(x) + g_3(x) = 0 \qquad \forall x, \tag{5.10}$$

so the rapid term conserves turbulent kinetic energy pointwise in wavenumber, not merely in the integral; and each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection $\kappa_a\kappa_b/\kappa^2$ degenerates. The signs are physically ordered: $g_1 > 0$ (energy fed to the streamwise component E_1), $g_2 < 0$ (energy removed from the line/upwash component E_2) over the bulk of the angular range, with g_3 changing sign, consistent with the plane strain stretching x_1 and compressing x_2.

For *isotropic* turbulence the anisotropy scalar vanishes ($c = 0$) and the isotropic-like scalar a depends only on κ. Averaging the kernels over solid angle—equivalently, integrating $g_n(x)$ against the isotropic distribution of x on the sphere—gives

$$\langle g_1\rangle : \langle g_2\rangle : \langle g_3\rangle = +\tfrac{1}{5} : -\tfrac{1}{5} : 0, \tag{5.11}$$

in the ratio $+1 : -1 : 0$. This is exactly the structure of the plane strain $S_{ij} = \mathrm{diag}(\tfrac{1}{2}, -\tfrac{1}{2}, 0)$: the rapid term is proportional to S_{ij}, with $\Pi_{33}^{(\mathrm{r})} = 0$ because the strain has no component along x_3. The spectral kernel therefore collapses, under isotropy, to the isotropic-production form of the moment closure,

$$\Pi_{ij}^{(\mathrm{r})} = \tfrac{4}{5}\, k_t\, S_{ij} \qquad \text{(isotropic limit)}, \tag{5.12}$$

the result of Crow (1968) that underlies the isotropisation-of-production (IP) rapid term in the closure of Launder *et al.* (1975). We note a convention: with the symmetrised definition of the pressure–strain correlation adopted here and the factor of two carried in the Poisson source (2.7), the kernels (5.7)–(5.9) yield the coefficient $\tfrac{2}{5}k_t$ for the 11-component directly (since $S_{11} = \tfrac{1}{2}$), equivalent to the standard $\tfrac{4}{5}k_t S_{ij}$ of (5.12); the ratio (5.11) and the tracelessness (5.10) are independent of this convention.

The consequence is that the algebraic closure used to obtain the distortion of § 4 is not an independent assumption but the isotropic-limit reduction of the line-consistent spectral term derived here. The spectral form (5.6) carries, in addition, the angular (x-dependent)

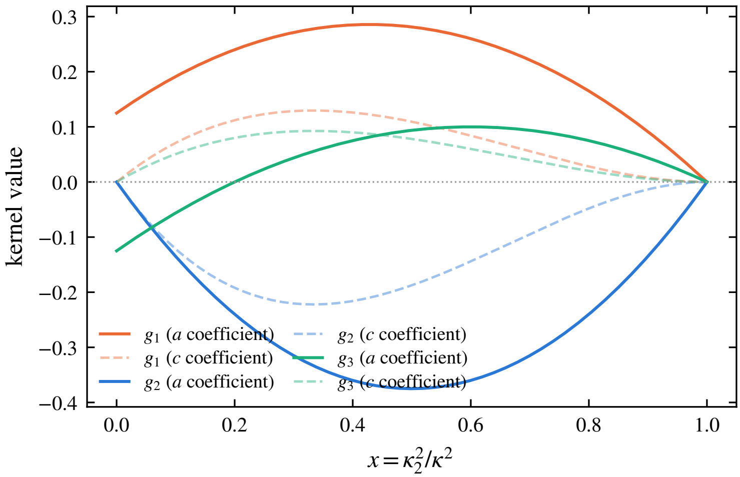

Figure 12. Rapid pressure–strain closure kernels g_1, g_2, g_3 (eqs. (5.7)–(5.9)) against the angular variable $x = \kappa_2^2/\kappa^2$. Solid curves: coefficient of the isotropic-like spectral scalar a; faint dashed curves: coefficient of the anisotropy scalar c. The kernels sum to zero at every x (energy conservation, eq. (5.10)) and vanish at $x = 1$. Averaged over an isotropic spectrum they give the ratio $+\frac{1}{5} : -\frac{1}{5} : 0$ (eq. (5.11)), recovering the isotropisation-of-production form $\Pi_{ij}^{(\mathrm{r})} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) that underlies the moment closure used in § 4.

and anisotropy (c-scalar) information that the moment closure discards—and both discards are now measured: the wavenumber dependence of the linear response in the three-way comparison of § 4.1.1, and the component ordering in § 5.6. The moment closure and the collapsed form thus agree to leading order, and the terms in which they depart are exactly the measured deficiencies of § 4.

5.8. *Consistency with the eddy redistribution: avoiding double counting*

A closed rapid term is necessary but not sufficient: it must not duplicate redistribution already effected by the eddy events. The pressure–strain correlation decomposes canonically into rapid and slow parts, $\Pi_{ij} = \Pi_{ij}^{(\mathrm{r})} + \Pi_{ij}^{(\mathrm{s})}$, corresponding to the two source terms in (2.7). The slow part $\Pi_{ij}^{(\mathrm{s})}$ is the return-to-isotropy mechanism, quadratic in the fluctuations and independent of the mean gradient; the eddy events in ODT, which act on the fluctuating field without reference to A_{kl}, are the model's representation of exactly this slow redistribution. The rapid part $\Pi_{ij}^{(\mathrm{r})}$, proportional to A_{kl} and linear in the fluctuations, is by construction absent from the eddy mechanism, which carries no dependence on the mean strain. The two therefore occupy disjoint terms of the canonical decomposition, and adding (5.6) as an explicit mean-strain-driven source alongside the eddies does not double-count, provided the eddy events are calibrated to reproduce the slow return-to-isotropy in the absence of mean strain—a condition that the no-strain baseline of § **??** verifies directly, since there the centroid relaxes under the eddies alone with no spurious rapid contribution. This argument establishes consistency at the level of the decomposition; the no-strain baseline of § **??** provides the direct numerical confirmation, the centroid there relaxing under the eddy mechanism alone.

The disjointness argument is now also a measured statement, from both sides. With the strain switched off the eddy events generate no spurious strain-like signal (the nominal-turbulence null of Appendix B). Conversely, in the rapid limit ($Sk_t/\varepsilon = 16$, under one eddy per line in $e \leq 1$) every kernel-energy allocation rule tested—the isotropic kernel, a strain-biased generalisation, and discrete eddy types—reproduces the exact rapid-distortion evolution of $b_{22}(e)$ to ±0.003: the rapid term cannot be reached from inside the eddy events, and the continuous operator carries it alone. Rapid and slow redistribution occupy disjoint terms by construction, and now by measurement.

5.9. *Summary: what the single-line treatment establishes*

1. The rapid term is exactly $A_{kl}M_{ijkl}$; its single-line representation is obstructed by an azimuthal null space that is stated exactly and survives the axisymmetric collapse at the level of the two spectral scalars.
2. Under A1 the term collapses to the kernel form (5.6)–(5.9); A1 is exact for axisymmetric distortion and is *measured* under plane strain, where its violation grows with accumulated strain and with strain rapidity (table 3).
3. Independently of A1's validity, line data bound the rapid term sharply; the bound is honest—and verified to bracket the exact term—up to $e \approx 0.25$–0.5 under plane strain, and at all computed strains under axisymmetric distortion.
4. The moment closure used in the simulations is the isotropic reduction of the collapsed form (verified, including the exact $\frac{4}{5}k_t S_{ij}$); what it discards is the wavenumber dependence of the linear response and the component ordering, both measured in § 4 and § 5.6.
5. Rapid and slow redistribution are disjoint: verified by the strain-off null and by the rapid-limit invariance of every kernel-energy allocation rule.
6. The acoustically relevant upwash amplification $b_{22}(e)$ is linear to $e = 1$ and robust to the strain geometry that controls A1's validity.

6. Measured limits, and scale-local relaxation

The envelope of § 4.1.2 localises the model's spectral deficiency in its wavenumber-uniform rapid kinematics: the strain-induced anisotropy is allocated uniformly across scales where linear theory and DNS concentrate it at the energy-containing range and remove it from the fine scales. This section asks the constructive question: which mechanism available to the model can supply the missing scale dependence? We answer it experimentally, by a scale-resolved diagnostic and a controlled family of interventions in the eddy mechanism, each tested at 1024 realisations with seeds paired to a common baseline, and each either falsified or confirmed by the same measurement. The outcome is a single measured statement: what is missing is *scale-local relaxation*, and supplying it—at the scales of the imbalance—is both necessary and, at moderate strain, sufficient to move the fine-scale anisotropy.

6.1. *A transmission diagnostic*

The configuration is a scale-diagnostic variant of the strained line: unit strain rate (so $e = t$), an initial spectrum peaked at a wavenumber κ_p resolved by more than a decade of smaller scales, and two measurement bands, $\kappa_2 \in [0.6, 2]\,\kappa_p$ (the energy-containing band) and $[6, 16]\,\kappa_p$ (roughly two triplet-map generations below it, since each map compresses by a factor of three). On each band the component imbalance is $\mathcal{A}_b = E_2^{\|}(b)/\frac{1}{2}[E_1^{\|}(b)+E_3^{\|}(b)]$, the band-integrated upwash energy over the transverse mean; all statistics are medians over realisations with bootstrap confidence intervals, for the intermittency reason given in § 4.1. The model's eddy and kernel mechanisms are component-symmetric, so $\mathcal{A} = 1$ is their fixed point; the strain drives $\mathcal{A}$ up at the scales where it deposits anisotropy, and the ratio $\mathcal{A}_{\mathrm{hi}}/\mathcal{A}_{\mathrm{lo}}$ measures how much of the large-scale imbalance survives transmission down the cascade.

The baseline (figure 13, black) exhibits the deficiency of § 4.1.1 in transmission form: $\mathcal{A}_{\mathrm{hi}}/\mathcal{A}_{\mathrm{lo}}$ stays near unity out to $e = 3$ (1.02, 0.95, 0.89 at $e = 1, 2, 3$)—the anisotropy injected at the energy-containing scales crosses two map generations essentially undiminished—and drops only at the deepest strain. The mechanism is direct: the high band is populated by the maps of *large* eddies (one generation takes κ_p to $3\kappa_p$, a second to $9\kappa_p$, inside the band), so fine-scale anisotropy is delivered by large-eddy events, not by a chain of small ones.

6.2. *Interventions: what fails and what works*

Table 4 summarises the intervention family. Two families fail for instructive reasons. Acceptance gating—rejecting accepted eddies whose post-event state remains anisotropic, at thresholds spanning 45 to 98 per cent rejection—is scale-blind: at mild thresholds the surviving eddy population reproduces the baseline anisotropy budget in every band (the gate self-selects eddies whose kernels were already going to do the relaxing, and the unrelaxed field keeps generating candidates), while at the harshest threshold the model merely drifts toward its eddy-free limit. Restricting the gate to sub-band-scale eddies sharpens the negative result into the mechanism statement above: removing up to 97 per cent of small-eddy events leaves both the fine-scale anisotropy *and* the fine-scale energy unchanged, so no acceptance-gating scheme can control transmission at this Reynolds number. Reweighting the triplet-map images (a middle image of relative size $\frac{2}{3}$, which halves the compression of the map's dominant central channel) moves one-point and energy-containing statistics—$\mathcal{A}_{\mathrm{lo}}$ falls significantly and the upwash energy fraction with it—but not the fine scales: with the image fractions summing to one, slowing the middle channel necessarily speeds the outer ones, and one outer image maps the energy peak directly into the measurement band.

What works is relaxation applied *at the scale of the imbalance*. After each accepted eddy's map and kernels, the same energy-equalising kernel procedure (§ 2) is applied to each of the three map images—the sub-interval's own triplet-map displacement shape as the kernel, coefficients from the sub-interval's own integrals, so that each sub-application conserves the momentum components and the total kinetic energy exactly, with no map applied and no random numbers drawn—and then recursively to 9 and 27 sub-regions. One level ($\ell/3$ for an eddy of size ℓ) is null: for the large eddies that feed the band, $\ell/3$ still sits above it. Two levels produce the first statistically significant fine-scale isotropisation of the family ($\mathcal{A}_{\mathrm{hi}}$ from 1.43 to 1.27 at $e = 1$, both bands' paired contrasts individually significant), and three levels extend the effect to deep strain: the levels that work, $\ell/9$ and $\ell/27$, are exactly the first that reach the measured band. The bookkeeping is clean throughout—band energies move by a few per cent, the upwash energy fraction and the intermittency tails are unchanged—and the pre-strain fine-scale floor moves toward component equality immediately ($\mathcal{A}_{\mathrm{hi}}$ from 0.83 to 0.95 at $e = 0$).

Two further variants separate the candidate explanations for the one remaining limitation, the fading of the gains beyond $e \approx 3$. Tying the recursion depth to the eddy size (subdividing while the sub-interval remains above a near-dissipative floor, so that large eddies do proportionally more sub-scale work) is the best member of the family: both bands significant at $e = 1$, the effect retaining its sign through $e = 3$, and no reversal at the deepest strain.

intervention (knob)	fine-scale effect	reading
acceptance gate on post-event anisotropy (45–98% rejection)	null at all thresholds; harshest drifts to eddy-free limit	gate is scale-blind: modulates eddy rate, not the scale allocation
gate on sub-band eddies only (42%, 97% rejection)	null, including band *energy*	fine scales are fed by large-eddy maps directly
unequal map images (middle $\frac{2}{3}$)	null (one-point statistics move: $\mathcal{A}_{\mathrm{lo}}$ 2.85 $\rightarrow$ 2.50 at $e = 3.9$)	outer images feed the band in one step; channels trade off
sub-scale kernels, 1 level ($\ell/3$)	null within CIs	$\ell/3$ sits above the measured band
sub-scale kernels, 2 levels ($\ell/9$)	$\mathcal{A}_{\mathrm{hi}}$: 1.43 $\rightarrow$ 1.27 at $e = 1$ (significant, both bands)	first significant fine-scale isotropisation
sub-scale kernels, 3 levels ($\ell/27$)	significant at $e = 1$ *and* $e = 3$ (-12% paired)	deepest levels reach the band
depth tied to eddy size	significant in both bands at $e = 1$; no late-strain reversal	best of the family: gains without the deep-strain overshoot
2 levels iterated 3× per event	strongest at $e = 1$ (1.29); overshoots at deep strain	per-event amplitude is not the limiter

Table 4. The intervention family on the transmission diagnostic (1024 paired-seed realisations per case; effects quoted as median paired same-seed contrasts, "significant" meaning the 95% bootstrap interval excludes zero).

Iterating the two-level pass three times within each event answers the amplitude hypothesis in the negative: it is the strongest early isotropiser but overshoots at deep strain. Depth that reaches the measured scales therefore matters; per-event dosage does not; and the deep-strain fade is universal across the family however the relaxation is administered. The remaining suspect is structural: relaxation is locked to eddy events while the strain resupplies the imbalance continuously, and at these total strains the event rate cannot keep pace. Whether that saturation is a model artefact or the physical statement that sustained rapid distortion outruns any cascade-mediated return to isotropy is not decided by these experiments.

6.3. *The measured missing ingredient*

The arc closes the loop opened by the three-way comparison. The deficiency measured there—anisotropy allocated uniformly over wavenumber—is confirmed here in transmission form, is unfixable by gating or map reweighting, and is corrected, at moderate strain, by exactly one ingredient: relaxation applied at the scale of the imbalance. At the formulation level this ingredient has a natural continuous home: the spectral kernel form (5.6)–(5.9) is precisely a wavenumber- and angle-dependent rapid operator, and replacing the wavenumber-uniform moment closure of § 2 by a discretisation of it is the formulation-level analogue of the sub-scale kernels' event-level correction. The present results bound what such an upgrade can and cannot deliver: it can re-allocate the rapid redistribution across scales (the measured deficit of § 4.1.1), and it cannot, any more than the event-level mechanisms, outrun sustained resupply at deep strain.

7. Consequence for leading-edge noise

[M5: detachable LE section — decision D1.]

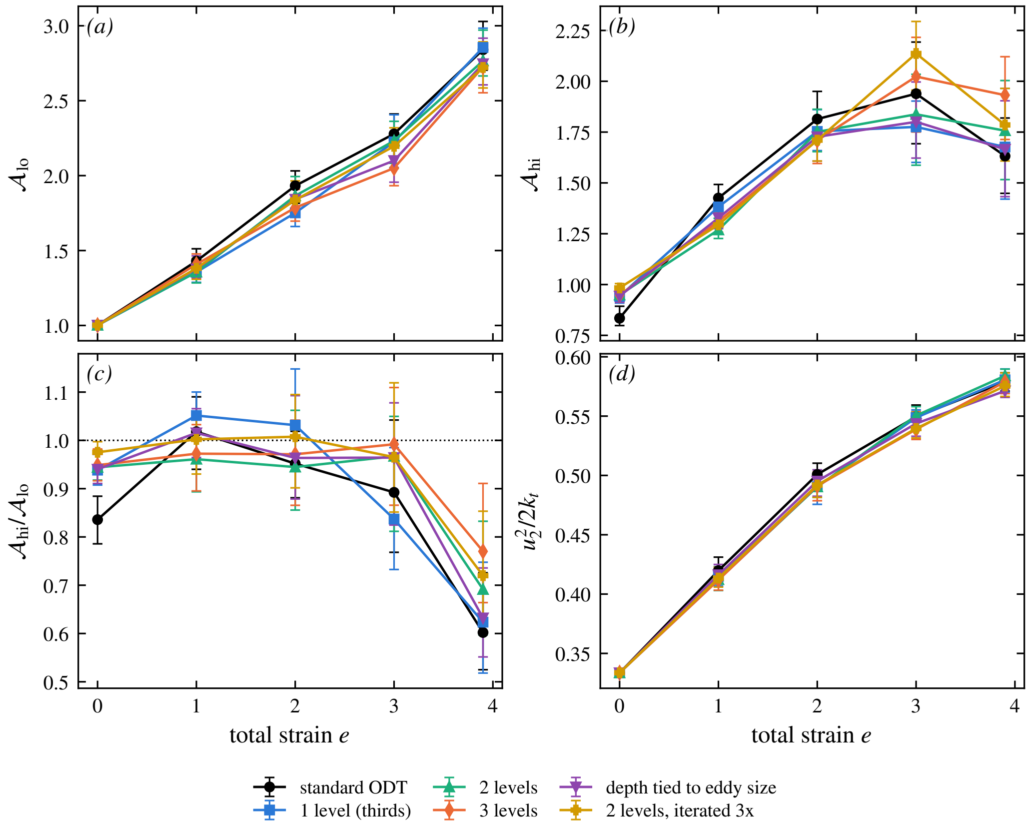

Figure 13. Scale-local relaxation on the transmission diagnostic: (*a*) energy-containing-band imbalance $\mathcal{A}_{\rm lo}$, (*b*) fine-scale-band imbalance $\mathcal{A}_{\rm hi}$, (*c*) transmission $\mathcal{A}_{\rm hi}/\mathcal{A}_{\rm lo}$, (*d*) upwash energy fraction, against total strain: standard ODT, hierarchical sub-scale kernels at fixed depth (1–3 levels), depth tied to eddy size, and the per-event iterated variant. Medians over 1024 paired-seed realisations, 95% bootstrap intervals.

8. Conclusions

[M4: conclusions rewrite.]

Appendix A. The forward map and the bound kernels

This appendix derives the three ingredients that § 5 uses: the two-scalar representation of the axisymmetric solenoidal spectrum with its realisability cone, the forward map from the two scalars to the line spectra (with the falsifiers T1–T2), and the closure kernels (5.7)–(5.9), together with the linear programs that turn these linear relations into the sharp bounds of § 5.4.

A.1. *The axisymmetric representation and realisability*

The spectrum tensor of statistically homogeneous turbulence is Hermitian, positive semi-definite at each $\boldsymbol{\kappa}$, and solenoidal, $\kappa_m \Phi_{mn}(\boldsymbol{\kappa}) = 0$; it therefore has rank at most two, supported on the plane normal to $\boldsymbol{\kappa}$. Under A1 (axisymmetry about $\boldsymbol{e}_2$, mirror-symmetric, helicity-free) it is fixed by two scalar functions of $(\kappa_2, \kappa_\perp)$ (Batchelor 1946; Chandrasekhar 1950; Lindborg 1995):

$$\Phi_{mn}(\boldsymbol{\kappa}) = a(\kappa_2, \kappa_\perp)\, P_{mn}(\boldsymbol{\kappa}) + c(\kappa_2, \kappa_\perp)\, \xi_m \xi_n, \qquad \boldsymbol{\xi} = \boldsymbol{e}_2 - \mu \frac{\boldsymbol{\kappa}}{\kappa}, \quad \mu = \frac{\kappa_2}{\kappa}, \tag{A 1}$$

with $P_{mn} = \delta_{mn} - \kappa_m\kappa_n/\kappa^2$ the solenoidal projector, so that $\kappa_m\xi_m = 0$ and $|\boldsymbol{\xi}|^2 = 1-\mu^2 = 1-x$ with $x = \kappa_2^2/\kappa^2$: solenoidality is exact by construction, which is how the third would-be scalar of a general axisymmetric tensor is eliminated. For a unit vector $\boldsymbol{v}$ normal to $\boldsymbol{\kappa}$, $\Phi_{mn}v_mv_n = a + c\,(\boldsymbol{\xi}\cdot\boldsymbol{v})^2$, so the two eigenvalues of the polarization matrix on that plane are a (attained for $\boldsymbol{v} \perp \boldsymbol{\xi}$) and $a + (1-x)\,c$ (for $\boldsymbol{v} \parallel \boldsymbol{\xi}$). Positive semi-definiteness is therefore the pointwise *realisability cone*

$$a(\kappa_2,\kappa_\perp) \;\geq\; 0, \qquad a(\kappa_2,\kappa_\perp) + (1-x)\,c(\kappa_2,\kappa_\perp) \;\geq\; 0, \tag{A 2}$$

quoted in § 5.4. Isotropy is the special case $c = 0$ with a a function of κ alone.

A.2. *The forward map to the line spectra*

Writing $\boldsymbol{\kappa} = (\kappa_\perp\cos\theta,\ \kappa_2,\ \kappa_\perp\sin\theta)$, the azimuthal moments at fixed $(\kappa_2,\kappa_\perp)$ are $\langle\kappa_1^2\rangle_\theta = \langle\kappa_3^2\rangle_\theta = \frac{1}{2}\kappa_\perp^2$, $\langle\kappa_1^4\rangle_\theta = \langle\kappa_3^4\rangle_\theta = \frac{3}{8}\kappa_\perp^4$, $\langle\kappa_1^2\kappa_3^2\rangle_\theta = \frac{1}{8}\kappa_\perp^4$, and every odd moment vanishes. Applied to (A 1) the diagonal weights average to

$$\langle P_{11}\rangle_\theta = \langle P_{33}\rangle_\theta = \tfrac{1}{2}(1+x), \quad P_{22} = 1-x, \quad \langle\xi_1^2\rangle_\theta = \langle\xi_3^2\rangle_\theta = \tfrac{1}{2}x(1-x), \quad \xi_2^2 = (1-x)^2, \tag{A 3}$$

and every mixed weight to zero. Substituting into the definition (5.3) of the line spectra, $\phi_{mn}(\kappa_2) = \int_0^\infty \langle\Phi_{mn}\rangle_\theta\, 2\pi\kappa_\perp\,\mathrm{d}\kappa_\perp$, gives the **forward map** from the two scalars to the data a line carries:

$$\phi_{11}(\kappa_2) = \phi_{33}(\kappa_2) = 2\pi\int_0^\infty\Big[\,a\,\tfrac{1+x}{2} + c\,\tfrac{x(1-x)}{2}\,\Big]\kappa_\perp\,\mathrm{d}\kappa_\perp, \tag{A 4}$$

$$\phi_{22}(\kappa_2) = 2\pi\int_0^\infty\Big[\,a\,(1-x) + c\,(1-x)^2\,\Big]\kappa_\perp\,\mathrm{d}\kappa_\perp, \tag{A 5}$$

$$\phi_{mn}(\kappa_2) = 0 \qquad (m \neq n). \tag{A 6}$$

Equations (A 4)–(A 6) contain the two free falsifiers quoted in § 5.4: under A1 the two transverse line spectra coincide at every resolved wavenumber (T1), and the cross-spectra vanish (T2); both are properties the line itself reports, scale by scale, with no reference to unresolved data. The ϕ_{22} bracket is the same combination of a and c that appears in the kernel g_2 below—the conventions mesh, as they must, because ϕ_{22} and the 22-component of the rapid term weight the spectrum through the same projection geometry.

They also explain why the isotropic intuition about line data fails under A1. For isotropic turbulence ($c = 0$, $a = a(\kappa)$) the map is over-determined: one unknown function of one variable against two measured functions, with ϕ_{22} determining a by the classical Abel-type inversion and the transverse spectrum slaved to it by the derivative relation. Under A1 the bookkeeping reverses: *two unknown functions of two variables* against two measured functions of one variable. At each fixed κ_2 the data supply exactly two numbers—the two weighted $\kappa_\perp$-integrals (A 4)–(A 5)—about two unknown functions of $\kappa_\perp$; any perturbation $(\delta a, \delta c)(\kappa_\perp)$ annihilating both integrals while respecting (A 2) is invisible to the line. This infinite-dimensional null space is the quantitative content of the obstruction of § 5.2 after the axisymmetric collapse.

A.3. *Derivation of the closure kernels*

For the diagonal components the exact result (5.2) reads

$$\Pi_{nn}^{(\mathrm{r})} = 2\,A_{kl}\int_{\mathbb{R}^3}\frac{\kappa_n\kappa_k}{\kappa^2}\,\Phi_{nl}(\boldsymbol{\kappa})\,\mathrm{d}\boldsymbol{\kappa} \qquad (\text{no sum on } n). \tag{A 7}$$

Substituting (A 1), carrying out the azimuthal average with the moments above, and specialising to $A = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ reduces each component to a $2\pi\kappa_\perp$-weighted integral over the half-plane of a polynomial combination in x, linear in (a, c)—the kernels (5.7)–(5.9) of the main text. The upwash component illustrates the mechanics: only $k = l = 1$ and $k = l = 2$ contribute, with azimuthal averages $\langle\kappa_1\Phi_{21}\rangle_\theta = -\frac{1}{2}\kappa_\perp^2\,\kappa_2\,[\,a + (1-x)\,c\,]/\kappa^2$ and $\langle\Phi_{22}\rangle_\theta = (1-x)[\,a + (1-x)\,c\,]$, so that both enter through the same bracket and

$$\Big\langle\text{integrand of (A 7)}\Big\rangle_\theta\Big|_{n=2} = 2\Big[-\tfrac{1}{2}\cdot\tfrac{x(1-x)}{2} - \tfrac{1}{2}\,x(1-x)\Big]\Big[a + (1-x)c\Big] = -\tfrac{3}{2}\,x\,(1-x)\Big[a + (1-x)c\Big], \tag{A 8}$$

which is $g_2(x)$. The transverse components involve in addition the fourth moments of (A 3)'s underlying table and yield g_1 and g_3; the full reduction has been verified symbolically against (A 7). Three identities check the algebra: $g_1 + g_2 + g_3 = 0$ at every x for both scalars (pointwise energy conservation); each kernel vanishes at $x = 1$, where the wavevector aligns with the line and the projection degenerates; and averaging over the isotropic angular measure $\mathrm{d}\mu$ on $[0,1]$ gives $\langle g_1\rangle : \langle g_2\rangle : \langle g_3\rangle = +\frac{1}{5} : -\frac{1}{5} : 0$, reproducing $\Pi^{(\mathrm{r})}_{ij} = \frac{4}{5} k_t S_{ij}$ (Crow 1968) as in § 5.7.

A.4. *The bounds as linear programs*

The target (5.6), the data constraints (A 4)–(A 5), and the cone (A 2) are all linear in (a, c), and everything decomposes over the resolved wavenumber: writing $\Pi^{(\mathrm{r})}_{nn} = \int_0^\infty \pi_n(\kappa_2)\,\mathrm{d}\kappa_2$ with $\pi_n = 2\pi \int_0^\infty g_n(x)[a,c]\,\kappa_\perp\,\mathrm{d}\kappa_\perp$, the extremisation over all spectra consistent with the line data is a family of *independent* infinite-dimensional linear programs, one per κ_2:

$$\pi_n^{\pm}(\kappa_2) = \max/\min_{(a,c)(\kappa_\perp)} \; 2\pi \int_0^\infty g_n(x)\,[a,c]\,\kappa_\perp\,\mathrm{d}\kappa_\perp \quad \text{s.t.} \quad \text{(A 4)–(A 5) hold, (A 2) pointwise,} \tag{A 9}$$

whence $\Pi^{\pm}_{nn} = \int \pi_n^{\pm}\,\mathrm{d}\kappa_2$ and the indeterminacy $\mathcal{I}_n = (\Pi^{+}_{nn} - \Pi^{-}_{nn})/2k_t S$ of § 5.4, normalised by the natural rapid-term scale so that $\mathcal{I}_n$ is comparable across strains (the isotropic value is $\Pi^{(\mathrm{r})}_{11} = \frac{2}{5} k_t$ per unit $S_{11} = \frac{1}{2}$). Discretised on a logarithmic $\kappa_\perp$ grid of order 10^2 points, (A 9) is a linear program with two equality constraints and the cone, solved per κ_2 in milliseconds; the bounds are global and sharp by LP duality—no fitting and no local minima. The side conditions of § 5.4 enter as further linear constraints: the log-Lipschitz slope cap $|\mathrm{d}\ln(a,c)/\mathrm{d}\ln\kappa_\perp| \le \lambda$ and the polarisation cap $|c| \le \gamma a$. The LP extremals concentrate on few-point supports in $\kappa_\perp$ (bang–bang spectra), which is why the raw bounds are conservative and why smoothness alone barely narrows them: a log-Lipschitz spectrum can still vary like $\kappa_\perp^{\pm\lambda}$, so the angular reweighting that drives the indeterminacy survives smoothing, while the polarisation cap attacks it directly. The dual variables of (A 9) certify, per resolved wavenumber, which linear combinations of the measured line spectra control each π_n; finite-window bias on ϕ_{mn} at small κ_2 propagates linearly and can be carried as interval data in place of the equalities with no change of machinery.

Appendix B. Numerical parameters and initialisation

[Appendix B: numerics + IC whitening + statistics policy.]

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